

# Rank Ultra-Log-Concavity of Preorder Gamma and Directed Support Polynomials at the Actual Degree

Degree four, independent role activities, small covers  
and universal-sink clouds

Research manuscripts prepared for Vladimir Reshetnikov  
(PDF metadata of sources 11, 12, 15, 16, 20, 22 and 25: “prepared with OpenAI”)  
merged in ProveIt’s batch-73 intake

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## Abstract

For a finite loopless directed relation  $D$ , for instance the off-diagonal relation of a finite preorder, let  $\gamma_k$  count the ordered pairs  $(S, T)$  of disjoint  $k$ -subsets of the ground set that admit a bijection  $S \rightarrow T$  along arcs, each pair counted once however many matchings witness it, and possibly weighted by independent tail and head activities. For a preorder,  $\Gamma_D = \sum_k \gamma_k t^k$  is the gamma polynomial of *preorder-root-polytopes*. That report asks (its Research question 96, `mr:q:gamma`) whether  $\Gamma_D$  is ultra-log-concave after normalization by its *actual* degree, and names degree three as the first open target. This report merges eleven manuscripts of ProveIt’s batch 73 that answer the question in three directions. Part I (sources 25, 16 and 23) proves it for every finite preorder of actual degree at most four with unit activities: degree three by a Gallai–Edmonds reduction and exact integer-population certificates, with the minimum order fifteen of a nonreal-rooted cubic; degree four by bounded cores, 76 four-attachment templates and 50 global binomial-square identities, with a smaller second route for the largest certificate. Part II (sources 20, 12, 11, 15 and 22) treats independent tail and head activities: real stability of the signed monomer polynomial under every two by two role cover, coefficientwise Rayleigh inequalities for a three-vertex shore, rank-ULC under every physical vertex cover of size three and on bipartite physical graphs, a pendant-head stability criterion, and rank-ULC of every independently weighted preorder of actual degree at most three. Part III (sources 14, 10, 34 and 06) treats an arbitrary directed core with a cloud of universal sinks: a sharp last-coefficient bound for every core size, rank-ULC at every actual degree for four-vertex cores, and for five-vertex cores at actual degrees five and four. The degree-three and degree-four theorems are computer-assisted; the structural theorems have ordinary proofs. Nothing here is formalized in Lean or Rocq, and nothing is refereed.

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# 1 Guide to this report

## 1.1 Sources, status and plan

This report merges research manuscripts delivered to ProveIt’s incoming-report intake on 1 October 2026 (batch 73, arrival commit `f8c3a392a`, placement commit `8f5c53106`). Sources are numbered by their batch-73 manuscript numbers; a package delivered in several archives takes the lowest number: source 16 is the four archives 16–19 (`preorder-degree-four`, parts 4 to 1), and source 06 is the two archives 06 and 07 (`universal-sink-five-core-boundary`, parts 2 and 1; its manuscript is in archive 07). Eighteen archives are behind the eleven printed sources. Two further archives are earlier versions of printed sources and are not sources: 31 (vertex activities, superseded by source 22) and 13 (a weaker four-core inequality, superseded by source 10). Appendix A records every archive.

The report has three Parts.

- Part I, unit activities on finite preorders through actual degree four: source 25 (degree three), source 16 (degree four, the base of the merge) and source 23 (a second route for the largest degree-four certificate).
- Part II, independent role activities and small covers: sources 20, 12, 11, 15 and 22.
- Part III, universal-sink clouds over an arbitrary directed core: sources 14, 10, 34 and 06.

Every source is printed with its statements, proofs, examples, remarks, limitations, verification notes and research questions; each source’s section opens with its abstract as delivered. A result proved by several sources is printed once, and the other proofs are kept as marked second routes or, where the proof is the same, replaced by a pointer recorded at the place. A result that *preorder-root-polytopes* or *matching-rank-normalization* already proves is cited by its label there and is not presented as new. Editorial additions are marked “[*Merge note*: ...]”.

*Status.* The sources are AI-assisted research manuscripts and notes: the title pages of sources 11, 12, 15, 16, 20, 22 and 25 read “Research manuscript prepared for Vladimir Reshetnikov” and their PDF metadata “Research manuscript prepared with OpenAI”; sources 06, 10, 14 and 34 read “Research note”; source 23 has no author line. Several main theorems are computer-assisted: their proofs include finite exact certificate computations shipped with this report (Section A.4). The report is unrefereed, and nothing in it is formalized in a proof assistant. No source claims literature-wide priority.

## 1.2 What the neighbouring reports already prove

Two reports of the collection precede this one. Their results are cited by label and are not reprinted. Line numbers are not used: the labels are stable.

- *preorder-root-polytopes* (in `enumerative-combinatorics/`). Theorem `gam:thm:main` expresses the  $h$ -polynomial of a finite preorder through the support numbers of this report, and Theorem `gam:thm:deficiency` says  $\deg \Gamma_\tau = \nu(\text{Comp}(\tau))$ . Theorem `lor:thm:weightedULC` proves two-shore ultra-log-concavity normalized by the smaller shore, and Corollary `lor:cor:gammaULC` all height-two gamma sequences. Theorems `ade:mat:thm:stable-bridge` and `ade:mat:thm:vertex-stable` treat stability of signed matching supports of bipartite graphs. In Part XII, Theorem `mr:thm:first` (with Lemma `mr:lem:MS`) is the universal first inequality for directed relations with non-negative activities, with a merge note crediting its vanishing-activity form; Theorem `mr:thm:rankthree` and Corollary `lc:cor:rankthree` settle bipartite matching rank three; Corollary `mr:cor:preorderfirst` gives the first inequality and hence every degree at most

two for preorder gamma polynomials; and Questions `mr:q:gamma` (Research question 96) and `mr:q:certificates` are the questions this report answers or bears on.

- *matching-rank-normalization* (beside this report). Corollary `mrn:w:rf:cor:rankfive`: every weighted bipartite graph of matching number at most five is ultra-log-concave of order its matching number. Theorem `mrn:w:sb:thm:main`: a bipartite graph with a vertex cover of two vertices on each shore has a real-stable support polynomial. Equation `mrn:s:tv:eq:rayleigh`: the scalar rank-three Rayleigh inequality (Wagner). Its Part V prints batch-73 manuscript 26 (source 77 there): Theorem `mrn:v:rf4:thm:main`, normalization by the actual degree through rank four, and Theorem `mrn:v:rf4:thm:mixed`, real roots under a two by two cover.

Sources 16 and 25 cite *preorder-root-polytopes* at its commit `d5e863bba`; the labels they cite are unchanged at the placement commit.

### 1.3 The common object

The following definitions are common to all eleven sources; each source restates them in its own words, and those restatements are kept in their sections because they fix each source's precise scope. Let  $D$  be a finite loopless directed relation on a set  $V$ . For a preorder,  $D$  is its off-diagonal relation: an arc  $i \rightarrow j$  means  $i \neq j$  and  $i$  precedes  $j$ , and equivalent elements give opposite arcs.

**Definition 1.1** (Feasible supports). A feasible support of size  $k$  is an ordered pair  $(S, T)$  of disjoint subsets of  $V$ , each of cardinality  $k$ , for which there exists a bijection  $\pi : S \rightarrow T$  with  $s \rightarrow \pi(s)$  for every  $s \in S$ . For nonnegative tail activities  $u_i$  and head activities  $v_i$  put

$$\gamma_k(D) = \sum_{(S,T) \text{ feasible, } |S|=k} u^S v^T, \quad u^S v^T = \prod_{i \in S} u_i \prod_{j \in T} v_j, \quad \Gamma_D(t) = \sum_{k \geq 0} \gamma_k(D) t^k.$$

The empty support contributes  $\gamma_0 = 1$ . A support contributes once, regardless of the number of witnessing bijections. With unit activities  $\gamma_k$  is the number of feasible supports of size  $k$ .

The same-ground-set disjointness is essential. Passing to independent tail and head copies without subsequently excluding their simultaneous use changes this polynomial. Counting individual matchings also changes it. A vertex may have different activities in its two roles; with zero activities the polynomial degree can decrease.

**Definition 1.2** (Rank-ULC). A polynomial  $g(t) = \sum_{k=0}^d g_k t^k$  is *rank-ultra-log-concave*, or rank-ULC, if its coefficients are nonnegative with no internal zeros and

$$\left( \frac{g_k}{\binom{d}{k}} \right)^2 \geq \frac{g_{k-1}}{\binom{d}{k-1}} \frac{g_{k+1}}{\binom{d}{k+1}} \quad (1 \leq k < d),$$

where  $d$  is the actual degree.

Normalization by the ground-set size, by a cover size, or by a presumed degree after some activities vanish is a different assertion, and Section 1.4 lists where a source uses one.

**Lemma 1.3** (Degree; sources 16, 20, 25). *For unit activities, the actual degree of  $\Gamma_D$  equals the matching number of the underlying undirected graph. Its coefficients have no internal zeros.*

*Proof.* A witnessing bijection yields an undirected matching. Conversely orient each edge of an undirected matching in a direction present in  $D$ . Its disjoint endpoint sets form a feasible support. Submatchings establish positivity at every smaller degree.  $\square$

[Merge note: for preorders this is *preorder-root-polytopes*, Theorem `gam:thm:deficiency`. Sources 16, 20 and 25 state and prove it as their Lemma “degree” with the proof above (source 25 for preorders, in its own words); source 22 adds a filtering argument for zero role activities, printed in Part II (Lemma of source 22). The statement is not new.]

**Lemma 1.4** (Motzkin–Straus bound; source 25). *If a simple graph  $H$  has clique number  $d \geq 1$  and nonnegative vertex weights  $w_v$ , then*

$$\sum_{uv \in E(H)} w_u w_v \leq \frac{d-1}{2d} \left( \sum_v w_v \right)^2.$$

*Proof.* Normalize the total weight to one, the zero-total case being immediate. If nonadjacent vertices  $u, v$  both have positive weight, keep their sum fixed and move all of it to the one with the larger weighted neighbor sum. The objective does not decrease, since it is affine in this division, and its positive support shrinks. Repetition leaves a clique of size  $q \leq d$ . On that clique the objective is  $(1 - \sum_v w_v^2)/2 \leq (1 - 1/q)/2 \leq (d-1)/(2d)$ . This is the classical compression argument of Motzkin and Straus [11].  $\square$

**Theorem 1.5** (Universal first inequality). *Let  $D$  be a finite loopless directed relation with nonnegative independent tail and head activities, and let  $d \geq 2$  be the actual degree of  $\Gamma_D$ . Then*

$$\gamma_1^2 \geq \frac{2d}{d-1} \gamma_2. \quad (1.1)$$

*Transitivity is not needed.*

*Proof.* Discard every arc of zero weight  $u_i v_j$ ; this changes no coefficient. Make a graph whose vertices are the remaining directed arcs and whose edges join arcs with four distinct physical endpoints. A clique is a positive-weight matching, and every positive-weight support has such a witness, so the clique number is exactly  $d$ . Give arc  $i \rightarrow j$  weight  $u_i v_j$ ; the total weight is  $\gamma_1$ . Every feasible support of size two is witnessed by at least one edge of this graph, and every witness has the prescribed support weight; a fixed pair of arcs determines only one ordered support. The weighted edge sum is consequently at least  $\gamma_2$ , and Lemma 1.4 bounds it by  $(d-1)\gamma_1^2/(2d)$ .  $\square$

[Merge note: this is *preorder-root-polytopes*, Theorem `mr:thm:first`, with the same proof; the form at the surviving degree for vanishing activities is credited in the merge note after that theorem. It is re-proved, by the same route and without novelty claim, as source 16’s Theorem “Universal first inequality”, source 25’s Proposition “first” (unit activities, with the lemma above), source 22’s Proposition “first”, source 11’s Lemma “first” and the first comparisons of sources 10, 34 and 06; sources 10, 11, 34 and 06 prove Lemma 1.4 by the same mass-shifting argument, and source 12 gives its triangle-free case for actual degree two. Those texts are replaced by pointers to this theorem. Source 25 adds, and we keep, that counting matchings instead of supports would change the second cubic gap, and that the comparison in this proof cannot be used backwards to prove it.]

In degrees two, three and four the inequality reads  $\gamma_1^2 \geq 4\gamma_2$ ,  $\gamma_1^2 \geq 3\gamma_2$  and  $3\gamma_1^2 \geq 8\gamma_2$ . In particular every  $\Gamma_D$  of actual degree at most two is rank-ULC and real-rooted; for preorders this is *preorder-root-polytopes*, Corollary `mr:cor:preorderfirst`.

## 1.4 Notions and notation

The eleven sources share one object but not one vocabulary. The following conventions hold throughout; every source section uses its own letters inside its own scope, and the table below says where a letter means different things.

| Notion                      | Meaning in this report                                                                                                                                                                        | Not to be confused with                                                                                                                                                                                                          |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Gamma_D, \gamma_k$        | ordered <i>disjoint</i> endpoint supports on the <i>same</i> ground set, each counted once (Definition 1.1)                                                                                   | the matching polynomial of $D$ , and the support polynomial of the two-copy bipartite enlargement; the latter is the $h$ -type setting of <i>preorder-root-polytopes</i> Parts XI–XIII and of <i>matching-rank-normalization</i> |
| Preorder gamma              | for a preorder $\tau$ , $\Gamma_\tau = \Gamma_D$ with $D$ its off-diagonal relation ( <i>preorder-root-polytopes</i> , <code>gam:thm:main</code> )                                            | the $h$ -polynomial $h_\tau = \sum_k \gamma_k t^k (1+t)^{n-2k}$                                                                                                                                                                  |
| Unit activities             | all $u_i = v_i = 1$ (Part I)                                                                                                                                                                  | “unweighted” in source 16 means unit activities; its “weighted quotient” means positive integer class sizes                                                                                                                      |
| Vertex activities           | one activity $x_i = u_i = v_i$ per vertex (archive 31, recorded as a corollary in Part II; source 23)                                                                                         | independent role activities                                                                                                                                                                                                      |
| Independent role activities | $u_i$ (tail) and $v_i$ (head), independent and nonnegative (sources 06, 10, 11, 12, 14, 15, 20, 22, 34)                                                                                       | a zero tail activity does not delete a head role (sources 06 and 22)                                                                                                                                                             |
| Actual degree               | $\deg \Gamma_D$ after zero activities are imposed                                                                                                                                             | the ground-set size, a cover size, or a degree presumed before activities vanish                                                                                                                                                 |
| Order- $r$ ULC              | log-concavity of $\gamma_k / \binom{r}{k}$ for a fixed $r \geq d$ (source 16’s exterior theorem, source 15’s shore order)                                                                     | rank-ULC; the two agree only when $r = d$                                                                                                                                                                                        |
| Role cover                  | sets $P$ (tails), $Q$ (heads) with every arc $i \rightarrow j$ having $i \in P$ or $j \in Q$ ; $P$ and $Q$ may overlap (sources 11, 15, 20, 22)                                               | a physical vertex cover of the underlying graph (source 11), and the bipartite cover of <i>matching-rank-normalization</i> with two vertices on each shore (the disjoint bipartite case)                                         |
| Bipartite physical graph    | the underlying undirected graph is bipartite; arcs may run both ways across (source 15)                                                                                                       | a bipartite <i>role</i> graph (tail and head copies), which every relation has                                                                                                                                                   |
| Template, attachment        | Gallai–Edmonds normal form: $a =  A $ attachments, exterior types with integer populations (sources 16, 23, 25)                                                                               | cores with exterior types in general; a universal-sink cloud is the case of one exterior type                                                                                                                                    |
| Rayleigh                    | coefficientwise (source 12, in formal activities); scalar on the nonnegative orthant (Wagner’s rank-three theorem, sources 16, 22, 23); all-real (the stability criterion, sources 20 and 22) | these are three different statements and are never interchanged                                                                                                                                                                  |
| Real stable                 | nonvanishing when all variables lie in $\mathbb{H} = \{\text{Im } z > 0\}$                                                                                                                    | the half-plane property (positive real parts), equivalent for homogeneous polynomials up to rotation (source 22)                                                                                                                 |

### Letters with several meanings.

- $P, Q$ : tail and head sets of a role cover in sources 15, 16, 20 and 22, and in source 11’s Corollary;  $Q$  is also a polynomial in sources 16, 22 and 23 ( $Q(t) = 1 + Lt + Bt^2 + Tt^3$ ,  $Q = \Gamma_{H-h}$ ), and  $P$  the polynomial  $\Gamma_H - Q$  in source 22. The report *preorder-root-polytopes* is never called  $P$  in the text.

- $A, B$ : in Part III (sources 06, 10, 34) the moment parameters  $A = \sqrt{p_2}$ ,  $B = p_1 - A$  of the sink activities; in source 22  $A_i$ ,  $B_{ij}$ ,  $C_0$  are three-core head sums; in sources 16 and 23  $A, B, C, D$  and  $T$  are coefficients of  $Q$  and  $R$ ; in source 16's role-cover material  $B_P, L_P$  are basis and element sums.
- $K$ : the core  $V \setminus I$  of the Gallai–Edmonds normal form in sources 16, 22 and 25; a number of two-element supports in source 23;  $K(r, m)$ , the sharp last-gap constant of source 14; the kernel  $p_0q_0 + p_1q_1$  in source 20's appendix.
- $D$ : the directed relation, and also the second cubic gap  $\gamma_2^2 - 3\gamma_1\gamma_3$  in source 25, the set of deficient vertices in the Gallai–Edmonds decomposition (sources 16, 22, 25), and a coefficient of  $R$  in source 16.
- $e_j$  and  $E_j$  (Part III): elementary symmetric functions of the core tail activities and of the sink head activities.  $e_j$  is the elementary symmetric polynomial itself.
- $\rho$ : the actual degree in the notation of sources 06, 10 and 34;  $d$  in the others.

No symbol of a source has been renamed. Where a source's normalization differs from the actual degree (source 16's Theorem on the exterior construction and source 15's shore order), the source says so at the place, and that sentence is kept.

**Labels and file names.** Labels of this report carry the prefix `pgr:`, with `pgr:u:`, `pgr:r:` and `pgr:s:` for Parts I, II and III and a two-letter infix per source (README). Each source refers to files of its delivered package by their delivered paths; the report README maps them to the shipped files in `code/` and `data/`.

## Part I

# Unit activities: finite preorders through actual degree four

This Part answers Research question `mr:q:gamma` of *preorder-root-polytopes* for unit activities through actual degree four. Section 2 (source 25) proves rank-ULC for every finite preorder of actual degree at most three, determines the least order of a preorder with a nonreal-rooted cubic gamma polynomial, and gives a general bounded-core normal form. Section 3 (source 16, the base of this report) proves rank-ULC for every finite preorder of actual degree four. Its two by two role-cover material is source 20's article and is printed once, in Part II. Section 4 (source 23) replaces the largest certificate of source 16 by a smaller certificate-assisted argument. Together:

**Theorem 1.6** (Sources 25 and 16). *Every finite preorder whose unweighted gamma polynomial has actual degree at most four is rank-ULC at that degree.*

The proof is Theorem 2.3 for degrees at most three and Theorem 3.9 for degree four (which quotes the former). Both are computer-assisted. Neither extends to activities other than one: Part II proves the independently weighted statement through degree three, and the weighted statement in degree four, and every statement in degree five or more, are open. Real-rootedness fails already in degree three (Section 2.9).

## 2 Degree at most three (source 25)

Source 25 is *Rank ultra log concavity for preorder gamma polynomials of degree at most three* (14 pages as delivered; archive `preorder-gamma-degree-three-package`). Its code, certificates and receipts are shipped with the prefix `25-degree-three-`.

**Abstract of source 25, as delivered.** We prove that the gamma polynomial of every finite preorder of actual degree at most three is ultra-log-concave after normalization by the binomial coefficients of that degree. The essential new case is the second cubic Newton inequality. A Gallai–Edmonds decomposition reduces arbitrary ground-set size to four cases: components of order at most seven; cores of order at most seven with one pendant population; cores of order at most five with two attachment vertices; and seventeen three-vertex-cover templates. Exact finite enumeration and rational polynomial certificates settle these cases for all nonnegative integer populations. The certificate package includes independent coverage and support-count checks, rather than only numerical tests of small instances. The distinction from real-rootedness is essential: an explicit fifteen-element height-two poset has gamma polynomial  $1+24z+162z^2+208z^3$  with negative discriminant. A separate exhaustive use of the same structural reduction proves that fifteen is the minimum order of such a nonreal-rooted cubic: every degree-three preorder on at most fourteen vertices is real-rooted, and at order fifteen the obstruction is unique up to isomorphism and order reversal. We also record a real-rooted infinite ordinal-sum family and describe the scope and reproducibility of the computation.

### 2.1 Statement and mathematical setting

A finite preorder  $\tau$  is a reflexive transitive relation on a finite set  $V$ . Its off-diagonal directed relation has an arc  $u \rightarrow v$  precisely when  $u \neq v$  and  $u \tau v$ . Antisymmetry is not assumed: distinct equivalent vertices give arcs in both directions. Write  $G = \text{Comp}(\tau)$  for the simple undirected graph with edge  $uv$  whenever at least one of  $u \rightarrow v$  and  $v \rightarrow u$  occurs.

**Definition 2.1** (Disjoint directed supports). A support of size  $k$  is an ordered pair  $(S, T)$  of disjoint subsets of  $V$ , with  $|S| = |T| = k$ , for which a bijection  $\pi : S \rightarrow T$  satisfies  $s \rightarrow \pi(s)$  for every  $s \in S$ . Let  $\gamma_k(\tau)$  be the number of these ordered pairs and put

$$\Gamma_\tau(z) = \sum_{k \geq 0} \gamma_k(\tau) z^k, \quad \gamma_0(\tau) = 1.$$

A support is counted once even when several bijections realize it.

This is the gamma polynomial identified in Part IV of the supplied root-polytope report [1], in particular its theorem labeled `gam:thm:main`. The present paper proves its statements directly for this explicitly defined polynomial. Thus the combinatorial result is independent of how that identification is implemented in the larger report. The convention differs both from counting ordinary matchings and from counting arbitrary row–column supports in two copies of the ground set: here the two endpoint sets must be disjoint in the original set.

[Merge note: source 25’s Lemma “degree” (for a preorder  $\tau$ :  $d = \deg \Gamma_\tau = \nu(G)$ , and all  $\gamma_k$  with  $0 \leq k \leq d$  are positive) is Lemma 1.3, printed once in Section 1.3 with the same proof; for preorders it is *preorder-root-polytopes*, Theorem `gam:thm:deficiency`.]

**Definition 2.2** (Normalization by actual degree). A polynomial  $g(z) = \sum_{k=0}^d g_k z^k$  with nonnegative coefficients and no internal zeros is *rank-ultra-log-concave*, or rank-ULC, if

$$\left( \frac{g_k}{\binom{d}{k}} \right)^2 \geq \frac{g_{k-1}}{\binom{d}{k-1}} \frac{g_{k+1}}{\binom{d}{k+1}} \quad (1 \leq k < d). \quad (2.1)$$

The parameter is the actual polynomial degree, not  $|V|$  or  $\lfloor |V|/2 \rfloor$ .

**Theorem 2.3** (Main theorem). *Every finite preorder  $\tau$  with  $\deg \Gamma_\tau \leq 3$  has a rank-ULC gamma polynomial. In actual degree three this says*

$$\gamma_1^2 \geq 3\gamma_2, \quad \gamma_2^2 \geq 3\gamma_1\gamma_3. \quad (2.2)$$

*In actual degree two the required inequality is the stronger  $\gamma_1^2 \geq 4\gamma_2$ .*

The proof has a classical structural input, elementary support arguments, and finite exact certificates. It is a computational proof with independently checked generators and identities; it is not a proof-assistant formalization. In particular, the conclusion quantifies over arbitrarily large finite preorders. No upper bound on the population sizes is used to prove Theorem 2.3.

The original report [1, Part XII, question `mr:q:gamma`] asks whether every preorder gamma polynomial is log-concave or rank-ULC and identifies degree three beyond the bipartite case as the first unresolved target. The theorem closes that degree-three target. It does not settle degree four or higher, a weighted version, or real-rootedness. A global literature-priority claim is not made.

[Merge note: this target is now closed further. Degree four with unit activities is source 16’s Theorem 3.9. The weighted version in degree three, with independent tail and head activities, is source 22’s Theorem (source 22, Part II); its case  $u_i = v_i = 1$  is Theorem 2.3, and the proof printed below, by integer-population certificates, and source 22’s proof, by a ledger of role certificates, are independent routes. Real-rootedness fails (Section 2.9).]

## 2.2 The universal first inequality

The first inequality is already established in the report [1, `mr:thm:first`]. Source 25 includes its short proof to make the normalization and logical dependencies explicit; that proof is printed once, in Section 1.3.

[Merge note: source 25’s Lemma “ms” (the Motzkin–Straus bound, with its compression proof) and Proposition “first” (for any finite off-diagonal directed relation with  $d = \nu(G) \geq 2$ ,  $\gamma_1^2 \geq \frac{2d}{d-1}\gamma_2$ ; transitivity is not needed) are Lemma 1.4 and the unit-weight case of Theorem 1.5, printed once in Section 1.3. Source 25’s proof of the proposition is the proof given there: different supports have disjoint sets of realizing pairs of arcs, so  $\gamma_2$  is at most the number of edges of the arc-disjointness graph.]

It follows immediately that all degree-two cases are rank-ULC and real-rooted, and that the first degree-three inequality always holds. Degrees zero and one have no internal inequalities. The remaining problem is precisely

$$D = \gamma_2^2 - 3\gamma_1\gamma_3 \geq 0 \quad \text{when } d = 3. \quad (2.3)$$

Counting matchings instead of supports would change  $D$ , and the comparison in the proof of Proposition 1.5 cannot be used backwards to prove this second inequality.

## 2.3 A canonical bounded core from Gallai and Edmonds

For any finite simple graph  $G$ , let  $D_G$  be the vertices omitted by at least one maximum matching, let  $A = N(D_G) \setminus D_G$ , and let  $C = V(G) \setminus (A \cup D_G)$ . The Gallai–Edmonds theorem states that the components  $D_i$  of  $G[D_G]$  are factor-critical,  $G[C]$  has a perfect matching, and every maximum matching matches the vertices of  $A$  into distinct  $D_i$ , with internal near-perfect matchings in all  $D_i$  and a perfect matching in  $C$ . We use the theorem in the form proved by West [17, Theorem 5]. In particular,

$$\nu(G) = |A| + \frac{|C|}{2} + \sum_i \frac{|D_i| - 1}{2}. \quad (2.4)$$

Assume from now on that  $\nu(G) = 3$ . Let  $I$  be the union of the singleton components of  $G[D_G]$  and put  $K = V(G) \setminus I$ . We call  $K$  the core. The set  $I$  is independent and  $N(I) \subseteq A$ . Write  $a = |A|$ ,  $|C| = 2c$ , and  $|D_i| = 2d_i + 1$  for each nonsingleton component, where  $d_i \geq 1$ . Equation (2.4) gives

$$a + c + \sum_i d_i = 3, \quad |K| = a + 2c + \sum_i (2d_i + 1) \leq 9 - 2a. \quad (2.5)$$

Here  $2d_i + 1 \leq 3d_i$  and  $2c \leq 3c$ . Hence  $a \in \{0, 1, 2, 3\}$ .

### 2.3.1 Why the preorder boundary has no equivalence ambiguity

An arbitrary vertex cover can cut through a preorder equivalence class. That would permit bidirectional exterior incidences and invalidate a fixed-sign exterior classification. The canonical choice above avoids this issue.

**Lemma 2.4.** *Every vertex of  $I$  is a singleton preorder equivalence class. At each attachment vertex  $v \in A$ , all arcs incident with nonisolated vertices of  $I$  have the same direction.*

*Proof.* Equivalent vertices  $u, v$  have identical relations to every third vertex by transitivity. In  $G$  they are adjacent true twins, so their transposition is an automorphism. The sets  $D_G, A, C$  are automorphism-invariant by their definitions. If an exterior vertex  $u \in I$  were equivalent to  $v \neq u$ , then  $v$  would also lie in  $D_G$  and be adjacent to  $u$ , contradicting that  $u$  is a singleton component of  $G[D_G]$ .

There are therefore no bidirectional arcs crossing from  $I$  to  $K$ . If distinct  $x, y \in I$  satisfied  $x \rightarrow v \rightarrow y$ , transitivity would force  $x \rightarrow y$ , contradicting independence. This excludes opposite directions at each fixed attachment.  $\square$

After discarding isolated exterior vertices, each exterior vertex is determined by a nonempty neighborhood  $T \subseteq A$ , together with the direction fixed at each member of  $T$ . There are at most  $2^a - 1$  possible types. Let  $n_T \in \mathbb{Z}_{\geq 0}$  be their populations.

**Lemma 2.5** (One-vertex compatibility is sufficient). *Fix a core preorder and a direction for each attachment. Keep precisely the nonempty neighborhood types for which adjoining one vertex gives a preorder. Then arbitrary nonnegative populations of all these types give a preorder.*

*Proof.* A transitivity failure has a witness  $u \rightarrow v \rightarrow w$  with  $u \not\rightarrow w$ . A witness involving at most one exterior vertex is already excluded by the core and single-type checks. A witness involving two distinct exterior vertices would have an attachment as its middle vertex and require both exterior directions there, which is forbidden. There are no exterior-to-exterior arcs, so these are all possibilities.  $\square$

Thus enlarging the family by allowing all nonnegative integer populations is safe. The canonical decomposition can impose additional neighborhood-surplus conditions, but there is no need to impose them for a positivity proof on the larger family.

| $a$ | Structural consequence                            | Verification needed                     |
|-----|---------------------------------------------------|-----------------------------------------|
| 0   | Every nontrivial component has at most 7 vertices | Finite real-rootedness lemma            |
| 1   | $ K  \leq 7$ , one pendant-twin population        | 2050 pairs of core polynomials          |
| 2   | $ K  \leq 5$ , at most three exterior types       | 1084 templates, 448 polynomial families |
| 3   | $K = A$ , $ K  = 3$ , at most seven types         | 17 templates                            |

### 2.3.2 A reusable normal form at every fixed degree

**Theorem 2.6** (General bounded-core normal form). *Let  $\tau$  be a finite preorder with comparability matching number  $r$ . The same canonical construction gives an attachment set  $A$ ,  $a = |A| \leq r$ , and an independent exterior  $I$  such that*

$$|K| \leq 3r - 2a, \quad \nu(G[K \setminus A]) = r - a.$$

*Every exterior point is a singleton equivalence class, every attachment has one fixed exterior direction, and there are at most  $2^a - 1$  nonisolated exterior types. The gamma coefficients have the finite expansion (2.7), of total population degree at most  $a$ .*

*Conversely, fix a core of size at most  $3r - 2a$ , a marked set  $A$  of size  $a$ , fixed directions, and the compatible exterior types. If  $\nu(G[K \setminus A]) \leq r - a$ , every population blow-up has matching number at most  $r$ . Thus all degree- $r$  preorders are represented by finitely many bounded-core polynomial families with unbounded nonnegative integer populations.*

*Proof.* Replace 3 by  $r$  in (2.4). Writing  $r = a + c + \sum d_i$  gives the same bound  $|K| \leq a + 3(c + \sum d_i) = 3r - 2a$ . The graph on  $K \setminus A$  is the disjoint union of  $G[C]$  and the nonsingleton factor-critical components, with matching number  $c + \sum d_i = r - a$ . Lemmas 2.4 and 2.5 do not depend on  $r$ . In any matching, each selected exterior vertex requires its own attachment, proving the quota bound and (2.7). For the converse, at most  $a$  matching edges touch  $A$  and every other edge lies in  $K \setminus A$ , so the matching size is at most  $a + \nu(G[K \setminus A]) \leq r$ . For fixed  $r$ , there are only finitely many core relations, marked sets, and direction choices.  $\square$

The restriction on  $\nu(G[K \setminus A])$  matters in the converse: the core-size bound alone does not imply matching rank at most  $r$ . The normal form reduces any fixed-rank coefficient inequality to finitely many population-polynomial families; it does not assert that arbitrary integer-polynomial positivity is decidable or that higher-degree ULC holds. At specializations where the actual degree drops, the normalization must be changed accordingly.

## 2.4 Finite enumeration and exact positivity methods

This section specifies what the computer checks and why the checks have the needed quantifiers. The data and source programs in the accompanying archive are part of the certificate.

### 2.4.1 Covering all finite preorder cores

Every finite preorder is uniquely a poset of equivalence classes with positive block sizes. Choose a linear extension of the quotient poset and label its  $q$  elements  $0, \dots, q-1$  in that order. Every strict comparison then points forward. We use two complete quotient enumerations:

1. Enumerate all subsets of the  $\binom{q}{2}$  forward pairs and keep the transitive relations.
2. Recursively add a last maximal element. Its predecessors must be an order ideal of the previous poset, and each such ideal gives one extension.

For every quotient, enumerate all positive ordered compositions of each allowed total core size into  $q$  block sizes. Expand each block to mutually equivalent vertices and each strict comparison to all corresponding directed arcs. Every preorder is represented up to relabeling. Repetitions are allowed; the recorded counts are not counts of isomorphism classes or of labeled preorders.

The naturally labeled quotient-poset counts for  $q = 1, \dots, 7$  are

$$1, 2, 7, 40, 357, 4824, 96428.$$

The expanded-core counts for total size  $n = 1, \dots, 7$  are

$$1, 3, 12, 68, 568, 7090, 131645. \quad (2.6)$$

For a fixed  $n$ , the count equals  $\sum_q Q_q \binom{n-1}{q-1}$ , where  $Q_q$  is the corresponding quotient count.

#### 2.4.2 Counting supports without counting matching multiplicities

For each ordered pair of disjoint equally sized endpoint sets, support feasibility is a Boolean perfect-matching test in the directed incidence relation  $S \rightarrow T$ . The package cross-checks three implementations: recursive matching search; Hall's condition for every subset of  $S$ ; and bottom-up construction of directed matchings with endpoint pairs stored in a set. The last method explicitly discards multiple realizations of the same support.

For a fixed core and exterior types, let  $q_T$  be the number of selected exterior vertices of type  $T$ . For fixed labeled representatives of those selected vertices, enumerate their source/target roles and the unused/source/target roles of core vertices; retain a role assignment exactly when a realizing matching exists. If this gives  $c_{k,q}$  feasible supports, then

$$\gamma_k(\mathbf{n}) = \sum_q c_{k,q} \prod_T \binom{n_T}{q_T}. \quad (2.7)$$

A support selects its exterior subsets uniquely, so this is an exact identity. Every used exterior vertex needs a different attachment in a realizing matching. Thus  $\sum_T q_T \leq a$ , making the sum finite regardless of the populations. In particular,  $\gamma_k$  has total population degree at most  $a$ , and  $\gamma_1$  has degree at most one.

Multiplication in the binomial basis can be done with integer arithmetic using

$$\binom{n}{a} \binom{n}{b} = \sum_{j=0}^{\min(a,b)} \frac{(a+b-j)!}{j!(a-j)!(b-j)!} \binom{n}{a+b-j}. \quad (2.8)$$

This counts pairs of subsets by the size  $j$  of their intersection. Taking products of this identity over the types gives exact binomial-basis expansions for  $D$ .

#### 2.4.3 Finite certificates for an infinite population domain

A polynomial with nonnegative coefficients in the multivariate binomial basis is nonnegative on  $\mathbb{Z}_{\geq 0}^r$ . For other polynomials we partition  $\mathbb{Z}_{\geq 0}^r$  according to the set  $F$  of positive coordinates and substitute

$$n_i = 0 \ (i \notin F), \quad n_i = 1 + x_i \ (i \in F), \quad x_i \geq 0. \quad (2.9)$$

Every nonnegative integer population vector lies on exactly one such face. A certificate is an exact identity expressing the substituted polynomial as a nonnegative rational combination of nonnegative monomials and terms  $x^m Z(x)^2$  with  $Z \in \mathbb{Q}[x]$ .

Some square terms are stored compactly as three monomials  $ux^\alpha + vx^\beta + wx^\eta$ , where  $u, v > 0$ ,  $w < 0$ ,  $\alpha + \beta = 2\eta$ , and  $w^2 = 4uv$ . Factoring  $x^{\min(\alpha, \beta)}$  leaves a square: the remaining endpoint exponents are even. All exponent vectors are nonnegative integers. General square terms include the rational polynomial  $Z$  as additional data. The verifier expands every term, checks every sign and exponent, and compares coefficients exactly.

Numerical linear programming and eigenvector-based column generation helped discover some certificates. They are not used to verify positivity. Verification contains no floating-point tolerance, approximate roots, or finite-population extrapolation. The face substitution proves an integer-population statement; it does not assert positivity for coordinates strictly between zero and one.

## 2.5 No attachments and one attachment

### 2.5.1 The finite real-rootedness lemma

**Lemma 2.7** (Exact finite lemma). *For every preorder on at most seven vertices,  $\Gamma_\tau$  has only real negative roots, with multiplicities allowed; the constant polynomial is included.*

*Computational proof.* The complete enumeration of Section 2.4 gives the core counts (2.6). There are no supports larger than three. A quadratic  $1 + bz + cz^2$  is real-rooted exactly when  $b^2 - 4c \geq 0$ . A cubic  $1 + bz + cz^2 + ez^3$ ,  $e > 0$ , is real-rooted exactly when

$$\Delta = b^2c^2 - 4c^3 - 4b^3e - 27e^2 + 18bce \geq 0. \quad (2.10)$$

Every enumerated coefficient vector passes the applicable exact integer test. Positivity of the coefficients excludes nonnegative roots. The generator and independent Hall checker are supplied in `code/small-check/`; representatives of all distinct gamma vectors are included. Direct enumeration of all reflexive relations through four vertices gives the known labeled-preorder counts 1, 4, 29, 355 and exactly the same coefficient sets. These checks supplement the coverage proof, rather than replace it.  $\square$

The literature already reports computer verification through seven vertices [2, Section 5]; Lemma 2.7 is independently reproduced here, not claimed as a new size range. The supplied final enumerator also verifies every degree-at-most-three preorder on eight vertices. It skips degree-four cases and is not evidence for a claim about their gamma polynomials.

If  $a = 0$ , there are no edges from  $D_G$  to  $C$  or between different  $D_i$ . The total order of  $C$  is at most six, and each nonsingleton  $D_i$  has at most seven vertices by the matching budget. Thus every nontrivial connected component has at most seven vertices. Support polynomials multiply over graph components: restricting a realizing matching gives equally sized tail/head sets in each component, and taking unions reverses this restriction. Lemma 2.7 therefore makes the whole polynomial real-rooted. Newton's inequalities imply rank-ULC with its total actual degree. This proves the  $a = 0$  branch without a separate ULC convolution argument.

### 2.5.2 One attachment and the support identity

Suppose  $a = 1$ , with attachment  $v$ . The core has at most seven vertices. After deleting isolated exterior vertices, every exterior vertex is a pendant neighbor of  $v$ , and all have the same direction. Reverse all arcs if needed to orient them  $v \rightarrow x$ . For  $m > 0$ , transitivity implies that no other core vertex has an arc into  $v$ , since  $u \rightarrow v \rightarrow x$  would force an additional neighbor of  $x$ . Thus  $v$  is a singleton minimal element of the core preorder.

If the exterior population is  $m$ , a support either avoids the exterior or uses exactly one exterior vertex. In the latter case the matching must use the pendant edge, and deletion of its endpoints gives a support in  $K - v$ . Conversely, every such support extends in  $m$  ways. Consequently

$$\Gamma_m(z) = \Gamma_K(z) + mz\Gamma_{K-v}(z). \quad (2.11)$$

If  $m > 0$ , a matching in  $K - v$  extends by a pendant edge, so  $\nu(K - v) \leq 2$ . The case  $m = 0$  is already Lemma 2.7.

Write  $\Gamma_K = 1 + p_1z + p_2z^2 + p_3z^3$  and  $\Gamma_{K-v} = 1 + q_1z + q_2z^2$ . The two cubic gaps in (2.11) are quadratics in  $m$  with coefficient triples  $(A, B, C)$ :

$$D_1(m) : \quad (1, 2p_1 - 3q_1, p_1^2 - 3p_2), \quad (2.12)$$

$$D_2(m) : \quad (q_1^2 - 3q_2, 2p_2q_1 - 3p_1q_2 - 3p_3, p_2^2 - 3p_1p_3). \quad (2.13)$$

Every certificate is either  $A, B, C \geq 0$ , or  $A > 0$  and  $4AC - B^2 \geq 0$ , using

$$4A(Am^2 + Bm + C) = (2Am + B)^2 + (4AC - B^2). \quad (2.14)$$

These prove nonnegativity for every real  $m \geq 0$ , a strengthening of what the preorder application requires.

The complete marked-core enumeration has 198469 valid marked expansions and 2050 distinct tuples  $(p_1, p_2, p_3, q_1, q_2)$ . By total core size the marked counts are

$$1, 3, 16, 115, 1160, 16786, 180388.$$

All 2050 first gaps and all 2050 second gaps pass: respectively 1114 and 1697 use nonnegative coefficients, and the remaining 936 and 353 use (2.14). Where  $p_3 = q_2 = 0$ , the actual degree can be two for all  $m$ ; all 136 such tuples also certify  $(p_1 + m)^2 - 4(p_2 + mq_1) \geq 0$ . At  $m = 0$  with  $p_3 = 0$ , the corresponding quadratic normalization is checked separately.

The independent C++ checker generates quotient posets by ideal extension and constructs supports by endpoint-mask dynamic programming. It reproduces every one of the 2050 tuple multiplicities, not merely one representative or the total count. A separate Python checker validates representatives and all 4236 stored quadratic certificates. This closes  $a = 1$ .

## 2.6 Two attachments

Let  $a = 2$ , so  $|K| \leq 5$ . Label the attachments  $v_0, v_1$ . Exterior neighborhoods are the nonempty subsets of this pair, encoded as masks 1, 2, 3. For every core preorder, pair of attachments, and two signs, the generator keeps exactly the types allowed by Lemma 2.5. It then identifies templates under permutations of the core respecting the attachment pair, interchange of the two attachments, and simultaneous reversal of all arcs.

Every resulting blow-up has matching number at most three, even when it is not a canonical Gallai–Edmonds instance. Indeed, a matching using  $r$  exterior edges has  $r \leq 2$  and size at most

$$r + \left\lfloor \frac{5-r}{2} \right\rfloor \leq 3.$$

There are at most three population variables. By (2.7), every gamma coefficient has population degree at most two and  $D$  has degree at most four.

**Proposition 2.8** (Exact two-attachment certificate). *For every core preorder of order at most five, every pair of attachment vertices, every fixed direction choice, and all nonnegative integer populations of the compatible exterior types,  $D \geq 0$ .*

*Computational proof.* The complete generator produces 1084 structural templates, with 5, 23, 141, 915 templates for core orders 2, 3, 4, 5, respectively. They yield 448 distinct full gamma-polynomial families. Grouping is used only when all coefficient polynomials agree in their ordered population variables. Of these, 284 have a coefficientwise nonnegative binomial expansion of  $D$ . Each of the other 164 is supplied with all its zero/positive face certificates, totaling 994 faces. Each face identity has the exact nonnegative form described in Section 2.4. Thus every nonnegative integer population vector is covered.

An independent audit enumerates all labeled preorders through core order five, obtaining 4, 29, 355, 6942 at orders 2, 3, 4, 5, respectively. It reproduces the complete set of 1084 canonical templates, reconstructs support polynomials using directed matchings deduplicated by endpoint sets, recomputes  $D$ , and checks every rational face identity. The audit does not call the generator's helper routines. An additional independent Hall implementation checks 19084 labeled blow-ups on the grid  $\{0, 1, 2, 3\}$  and recovers the binomial coefficients by finite differences; the known degree bound makes this polynomial reconstruction exact. This is an additional support-count check, not the proof of positivity on the unbounded domain.  $\square$

The files `templates.json`, `gap2_certificates.json`, the exact verifier, and both independent checking routes are in `code/two-attachment/`. A template with no allowable exterior type contributes only its small core and is covered by Lemma 2.7; omitting it from the nonempty-type export loses no case. If a specialization has actual degree at most two, Proposition 1.5 supplies the correct degree-two normalization. Hence Proposition 2.8 closes the  $a = 2$  branch.

## 2.7 Three attachments and seventeen templates

For  $a = 3$ , equation (2.4) forces  $C = \emptyset$  and every  $D_i$  to be a singleton. Thus  $K = A$  consists of exactly three vertices. The canonical exterior has singleton equivalence classes by Lemma 2.4. Every attachment has an exterior neighbor because a maximum matching matches all of  $A$  into the singleton  $D_i$ .

There are 29 labeled three-element preorders. Test all eight active sign vectors and all seven nonempty neighborhood masks. Requiring every attachment to have an allowed neighbor gives 130 labeled templates and 17 classes under core permutations and order reversal. The full list is below. A minus sign denotes core-to-exterior arcs, a plus sign exterior-to-core arcs. Mask  $t$  encodes the subset of  $\{0, 1, 2\}$  whose bits are one. The listed core arcs exclude loops and include transitive arcs.

| Class | Core arcs              | Signs | Allowed masks       |
|-------|------------------------|-------|---------------------|
| 0     | $\emptyset$            | ---   | 1, 2, 3, 4, 5, 6, 7 |
| 1     | $\emptyset$            | --+   | 1, 2, 3, 4          |
| 2     | 01                     | ---   | 1, 3, 4, 5, 7       |
| 3     | 01                     | --+   | 1, 3, 4             |
| 4     | 01                     | -+-   | 1, 2, 3, 4, 5       |
| 5     | 01, 02                 | ---   | 1, 3, 5, 7          |
| 6     | 01, 02                 | --+   | 1, 3, 4, 5          |
| 7     | 01, 02                 | -++   | 1, 2, 3, 4, 5, 6, 7 |
| 8     | 01, 02                 | +++   | 2, 4, 6, 7          |
| 9     | 01, 10                 | ---   | 3, 4, 7             |
| 10    | 01, 10                 | --+   | 3, 4                |
| 11    | 01, 02, 12             | ---   | 1, 3, 7             |
| 12    | 01, 02, 12             | --+   | 1, 3, 4, 5, 7       |
| 13    | 01, 02, 10, 12         | ---   | 3, 7                |
| 14    | 01, 02, 10, 12         | --+   | 3, 4, 7             |
| 15    | 01, 02, 10, 12         | +++   | 4, 7                |
| 16    | 01, 02, 10, 12, 20, 21 | ---   | 7                   |

### 2.7.1 The bipartite template

Class 0 is a bipartite graph with a source shore of size three and a target shore consisting of the exterior. Its directed disjoint supports are precisely the ordinary bipartite matching supports. The ambient-binomial normalization theorem of Röhrle and Ulirsch [13, Theorem A and Example 2.4] applies: for a relation on disjoint shores of sizes  $m, n$ , the support numbers normalized by  $\binom{\min(m, n)}{k}$  are log-concave. If the actual degree is three here, the smaller shore has size three, so this is exactly  $D \geq 0$ . Lower actual degrees are already settled by Proposition 1.5. [*Merge note:* class 0 is also a case of *preorder-root-polytopes*, Theorem `lor:thm:weightedULC` (two-shore ultra-log-concavity normalized by the smaller shore), proved there with the same transversal-matroid device.]

For clarity, the relation support numbers fit the published theorem by assigning an independent indeterminate to each edge of the incidence matrix. A minor is nonzero precisely when its row and column sets support a matching: different matching terms have different monomials and

cannot cancel. These minors define a representable bimatroid. No theorem about normalizing an arbitrary bimatroid by its actual rank is being invoked. This published ambient-normalization theorem is the only non-elementary polynomial-inequality input for the seventeen-template step.

### 2.7.2 Compact product and extension certificates

In classes 1, 3, 10, the polynomial factors as

$$\Gamma(z) = (1 + Az + Bz^2)(1 + Cz), \quad C = n_4.$$

Consequently

$$D = (B - AC/2)^2 + \frac{3}{4}C^2(A^2 - 4B). \quad (2.15)$$

The remaining expressions are

$$\begin{aligned} \text{class 1: } \quad & A = n_1 + n_2 + 2n_3, \quad B = n_1n_2 + (n_1 + n_2)n_3 + \binom{n_3}{2}, \\ & A^2 - 4B = (n_1 - n_2)^2 + 2n_3^2 + 2n_3; \\ \text{class 3: } \quad & A = n_1 + 2n_3 + 1, \quad B = n_1n_3 + \binom{n_3}{2}, \\ & A^2 - 4B = n_1^2 + 2n_1 + 2n_3^2 + 6n_3 + 1; \\ \text{class 10: } \quad & A = 2n_3 + 2, \quad B = \binom{n_3}{2}, \quad A^2 - 4B = 2n_3^2 + 10n_3 + 4. \end{aligned}$$

These are nonnegative on the required domain. For class 9, subtract the class-10 gap. All coefficients of the difference are nonnegative except possibly those in the pure  $n_7$  contribution. That whole univariate contribution is exactly

$$4\binom{n_7}{1} + 41\binom{n_7}{2} + 57\binom{n_7}{3} + 18\binom{n_7}{4}, \quad (2.16)$$

which is nonnegative for integer  $n_7 \geq 0$ . The coefficient assertion and identity are verified exactly in `verify_cover3_compact.py`. Classes 11, 13, 15, 16 have nonnegative multivariate-binomial expansions of  $D$  directly.

### 2.7.3 The complete face certificates

Classes 5, 6, 8, 12, 14 have a certificate for every zero/positive face using monomials and monomial-weighted binomial squares. Class 2 has such certificates on 31 of its 32 faces. The remaining face has only  $n_3, n_4$  positive; its gap is exactly that of class 3 at  $n_1 = 0$ , so (2.15) applies.

Classes 4 and 7 need some general rational square polynomials as well. Class 4 has 23 binomial-square face certificates and 9 general-square certificates, covering all 32 faces. Class 7 has  $73 + 55 = 128$  certificates, covering all its faces. The hardest saved class-7 identity has 210 nonzero square/monomial weights; the checker expands its 330 coefficient equations over the rationals.

The independent script `audit_cover3.py` separately enumerates the 29 core preorders, reproduces the 130 signed templates and all 17 canonical keys, and reconstructs every coefficient in (2.7) by Hall's condition on ordered endpoint masks. It recomputes every gap from the gamma coefficients and verifies the face partitions, missing-face reductions, exponent domains, and duplicate guards. The three exact identity verifiers then check all saved rational identities. Thus the certificates are linked to the intended support counts, not merely to an unchecked polynomial export. This proves  $D \geq 0$  throughout the  $a = 3$  branch.

## 2.8 Completion of the main theorem

*Proof of Theorem 2.3.* Lemma 1.3 identifies the degree and excludes internal zeros. Degrees zero and one are immediate; degree two follows from Proposition 1.5. In degree three the same proposition proves the first inequality. Apply the Gallai–Edmonds decomposition to the comparability graph. Equation (2.5) gives exactly four possibilities. The  $a = 0$  branch follows from the finite real-rootedness lemma and component multiplication; the  $a = 1$  branch from (2.11) and its complete quadratic certificates; the  $a = 2$  branch from Proposition 2.8; and the  $a = 3$  branch from Section 2.7. These prove the second inequality in every case. All population sizes are unrestricted nonnegative integers throughout these four arguments.  $\square$

The reduction is essential to the quantifier. A finite scan of all preorders up to a fixed number of vertices would not prove the theorem. Here the graph structure bounds the core, and algebraic certificates handle the unbounded exterior populations. The finite base computation for  $a = 0$  is sufficient because each component is provably small, even though isolated components may occur in arbitrarily large number.

## 2.9 Real roots and an explicit obstruction

Rank-ULC is strictly weaker than real-rootedness even in actual degree three. Take three minimal elements  $u_0, u_1, u_2$  and three disjoint four-element sets  $X, Y, Z$  of maximal elements. Let the strict relations be exactly

$$u_0 < X \cup Y, \quad u_1 < X \cup Z, \quad u_2 < Y \cup Z.$$

This is a height-two poset on fifteen vertices. Its three minimal vertices form a vertex cover, and a matching saturating them exists, so the actual degree is three.

There are 24 arcs. For each choice of two minimal vertices, all pairs of maximal vertices are feasible except pairs contained in either of the two four-element types that see only one chosen minimal vertex. Thus

$$\gamma_2 = 3 \left( \binom{12}{2} - 2 \binom{4}{2} \right) = 162.$$

A triple of maximal vertices fails to match all three minima exactly when all three lie in the same four-element type. Hall’s condition has no other obstruction because every maximal vertex sees two minima. Hence

$$\gamma_3 = \binom{12}{3} - 3 \binom{4}{3} = 208.$$

Therefore

$$\Gamma(z) = 1 + 24z + 162z^2 + 208z^3, \quad \Delta = -2592 < 0. \quad (2.17)$$

Its two cubic Newton gaps are 90 and 11268, both positive, consistent with Theorem 2.3. A direct endpoint-support enumerator independently reproduces (2.17); no population formula is used in that check.

[Merge note: the same example, with the same numbers, is computed in source 22 (Section “A cubic with nonreal zeros”, where it is noted that it lies in the pure three-tail family of source 22’s three-core theorem and needs no role asymmetry), in source 16 (“The preceding research ... exhibited the nonreal-rooted cubic”) and in archive 31. It is printed once, here.]

### 2.9.1 The exact minimum order in degree three

**Theorem 2.9** (Minimum order and uniqueness). *Every finite preorder on at most fourteen vertices whose gamma polynomial has actual degree three is real-rooted. Among fifteen-vertex*

*preorders of actual degree three with a nonreal gamma zero, there are exactly two ordinary isomorphism classes, represented by the height-two poset in (2.17) and its dual. Equivalently, there is one class modulo isomorphism and global order reversal.*

*Computer-assisted proof with global coverage.* Apply the canonical four-branch reduction. The  $a = 0$  branch is real-rooted at every allowed order by Lemma 2.7 and component multiplication. For  $a = 1, 2, 3$ , enumerate every population specialization whose effective total order is at most fifteen and whose cubic coefficient is positive. Test the exact integer discriminant (2.10). The complete counts are

| Branch            | Evaluations | Actual cubics | Negative discriminants |
|-------------------|-------------|---------------|------------------------|
| One attachment    | 19430       | 17785         | 0                      |
| Two attachments   | 84191       | 58800         | 0                      |
| Three attachments | 128726      | 121510        | 1                      |

The sole negative case has total order fifteen, class 0 of Section 2.7, and populations  $(n_1, \dots, n_7) = (0, 0, 4, 0, 4, 4, 0)$ . It is exactly (2.17). An independently written scanner regenerates every per-size count and the complete negative-case list using arbitrary-precision integers.

There are two important coverage points. In the two-attachment scan we retain all 1084 structural templates and their actual core sizes, not only the 448 polynomial families. In the one-attachment scan we group by  $(p_1, p_2, p_3, q_1, q_2)$  but retain a *minimum-order* realization of every tuple. The complete marked-core generator explicitly minimizes this size over all occurrences; all 2050 stored minima are verified by a fresh enumeration. Replacing a core by this minimum-size representative preserves (2.11) and can only reduce total order. Hence any example through a bound appears at some scanned order through that bound. Isolated vertices may be deleted or added without changing gamma. The displayed counts are therefore representation counts, not a per-order census of polynomials or preorder isomorphism classes.

For uniqueness, any nonreal cubic on at most fifteen vertices must lie in the canonical  $a = 3$  branch, after deleting isolates. Up to a permutation of the three core vertices and possibly global duality, its core, signs, and populations occur in the complete seventeen-template scan. The sole negative specialization has effective order fifteen, so no isolates were deleted. Its populations determine the directed relation up to permutations within types. Thus the preorder is the displayed poset or its dual. They are not isomorphic: one has three minimal and twelve maximal elements, and its dual has twelve minimal and three maximal elements.  $\square$

[*Merge note:* this theorem supplies the minimality statement that source 22 explicitly does not make (“No minimal-size claim is made”). It does not conflict with the nonreal gamma polynomials of *preorder-root-polytopes*: the eight-vertex example  $\Theta_3$  there has gamma degree four, and the only degree-three member of that family is real-rooted.]

This theorem concerns actual degree three only. It makes no claim about higher-degree preorders on at most fourteen vertices. Its bounded discriminant scan is logically separate from the all-population second-gap certificates used for Theorem 2.3. The scripts and receipts are in `code/minimum/`.

## 2.10 An infinite family with three negative roots

A complementary positive result holds for a large antichain whose complement has three vertices. Let  $\mathcal{A}_m$  be an antichain of size  $m$ , and let  $\oplus$  denote ordinal sum, with all elements of an earlier summand strictly below all elements of a later one.

**Theorem 2.10.** *If  $m \geq 3$  and  $P, Q$  are arbitrary finite preorders with  $|P| + |Q| = 3$ , allowing either to be empty, then  $\Gamma_{P \oplus \mathcal{A}_m \oplus Q}$  has degree three and three distinct negative real roots. Every ordinal sum of antichains of actual gamma degree three also has three distinct negative real roots.*

*Proof.* Every edge meets the three-element complement of  $\mathcal{A}_m$ , and its vertices can be matched to three distinct antichain vertices, so the degree is three. Reversal reduces the possible splits to two forms.

For  $\mathcal{A}_m \oplus R$ ,  $|R| = 3$ , let  $e$  be the number of off-diagonal arcs in  $R$  and let  $t$  be the number of its vertices with at least one outgoing off-diagonal arc. Then

$$(\gamma_1, \gamma_2, \gamma_3) = \left( 3m + e, 3 \binom{m}{2} + tm, \binom{m}{3} \right). \quad (2.18)$$

The feasible  $(e, t)$  are  $(0, 0), (1, 1), (2, 1), (2, 2), (3, 2), (4, 2), (4, 3), (6, 3)$ . For  $1 \oplus \mathcal{A}_m \oplus R_2$ ,  $|R_2| = 2$ , let  $j \in \{0, 1, 2\}$  be its number of off-diagonal arcs. Then

$$(\gamma_1, \gamma_2, \gamma_3) = \left( 3m + 2 + j, 5 \binom{m}{2} + (j + 1)m, 3 \binom{m}{3} \right). \quad (2.19)$$

To justify (2.18), a size-two support with two antichain points chooses two of the three core heads, while one with a single antichain point chooses a core tail with an outgoing arc. A size-three support uses all three core heads. For (2.19), two chosen antichain points admit five size-two role patterns: one with both as tails, and four with one tail and one head. One antichain point gives  $j + 1$  patterns, and three give three choices for the head matched from the lower singleton. These are Boolean support counts.

For all eleven forms, put  $x = m - 3$ . The coefficients of  $12\Delta/(x+3)$  are listed in Appendix 2.12; every one is positive. Thus  $\Delta > 0$  for  $m \geq 3$ . The cubic has three distinct real roots and, by positive coefficients, they are negative.

For an ordinal sum of antichains with total size  $n$  and maximum layer size  $M$ , its comparability graph is complete multipartite and has matching number  $\min(\lfloor n/2 \rfloor, n - M)$ . If  $M \geq n/2$ , match all outside vertices into the largest layer. If  $M < n/2$ , pairing vertices from two largest remaining parts constructs a matching missing at most one vertex; the invariant that no part exceeds the ceiling of half the remaining total is preserved. Degree three consequently means either  $n \in \{6, 7\}$  and  $M \leq n - 3$ , or  $n \geq 8$  and  $M = n - 3$ . Whenever  $M = n - 3$ , the preceding result applies. The remaining compositions have  $n = 6$  with all layers at most two, or  $n = 7$  with all layers at most three: respectively 13 and 44 cases. Their exact cubic discriminants are positive in the supplied table `layered_certificates.json`. An independent layer-quota recurrence and direct support enumeration verify all 57 cases.  $\square$

Two specializations are already in the literature. The two-antichain case  $(e, t) = (0, 0)$  is covered by Athanasiadis and Chapoton [2, Proposition 9.2]. The case  $(e, t) = (6, 3)$ , a three-element equivalence block above singleton leaves, is covered by Athanasiadis, Xiao, and Yan [3, Remark 2.3]. Neither specialization is claimed as new. The eleven-template statement is included as a proved complementary family; a complete literature-priority audit for its precise scope has not been performed.

## 2.11 Reproducibility and scope

The archive contains the editable LaTeX source, this PDF, exact source code, certificate data, checking receipts, and a checksum manifest. It contains no required compiled binaries or numerical-optimizer state. The top-level `README.md` gives the dependency versions and commands. The

primary checks need Python 3 with SymPy and a C++17 compiler. No optimization package is required to verify the saved certificates.

The command `python3 verify_all.py` runs the exact checking routes and writes fresh logs under `receipts/`. It checks the finite-core lemma; the full marked-core enumeration and all one-population quadratic identities; independent two-attachment coverage and all its rational certificates; the seventeen three-cover templates and their exact identities; the ordinal-sum family; the bounded scans proving Theorem 2.9; and the explicit nonreal-root example. The finite generators can also be rerun separately using the commands in the README. The main theorem depends on the published Gallai–Edmonds theorem and the class-0 ambient-normalization theorem as clearly stated in the text, not on a numerical solver.

[Merge note: file names in this section are those of source 25’s delivered package; the files are shipped as `code/25-degree-three-...` and `data/25-degree-three-...`, and the README maps the names and gives rerun instructions. At intake every step of `verify_all.py` reproduced its recorded outputs (on a copy, compared modulo line endings and timing fields) except the order-eight census of `small-check/exhaust`, whose optimized build crashed after finishing its enumeration and was not rerun; the C++ sources also need `-include cassert` with GCC 16. Of the research directions below, the first is answered by source 16 (Section 3) and the fourth, in degree three, by source 22 (Part II); the second, third and fifth remain open.]

The following directions remain meaningful beyond this result.

1. **Actual degree four.** The same matching-budget argument gives bounded canonical cores, but more exterior types and higher-degree inequalities. A direct certificate approach must control that growth rather than infer higher-rank behavior from the cubic case.
2. **A conceptual second-gap proof.** The rational certificates may reveal common decompositions that replace the finite case analysis. At present the general cubic second inequality genuinely uses the supplied exact certificate computation.
3. **Equality in the second inequality.** The first-gap equality theory in the original report is not automatically a classification for the second gap. Extracting and reconciling the zero sets of the certificates is a separate problem.
4. **Weighted variants.** The present theorem uses unweighted supports. Replacing populations by arbitrary real activities changes the domain and the coefficient model; the integer-face certificates do not by themselves establish such a theorem.
5. **Formal verification.** The small support kernels and rational identities are natural candidates for a proof-assistant checker. Exact arithmetic and independent implementations are valuable checks, but are not a formalized proof.

## 2.12 Explicit discriminant certificates for the ordinal family

For (2.18) and (2.19), the table lists the coefficients in ascending powers of  $x$  of  $12\Delta/(x+3)$ , with  $x = m - 3$ . Thus each row gives a standalone integer positivity certificate. The common factor  $x + 3$  is positive for  $x \geq 0$ .

| Form   | Parameters | $x^0$  | $x^1$  | $x^2$ | $x^3$ | $x^4$ | $x^5$ |
|--------|------------|--------|--------|-------|-------|-------|-------|
| Outer  | (0, 0)     | 8640   | 13248  | 7884  | 2286  | 324   | 18    |
| Outer  | (1, 1)     | 22484  | 30336  | 15577 | 3789  | 432   | 18    |
| Outer  | (2, 1)     | 30148  | 37836  | 17888 | 4014  | 432   | 18    |
| Middle | $j = 0$    | 41412  | 46188  | 18456 | 3378  | 288   | 6     |
| Outer  | (2, 2)     | 45376  | 55296  | 25472 | 5490  | 540   | 18    |
| Outer  | (3, 2)     | 60804  | 69480  | 29613 | 5877  | 540   | 18    |
| Middle | $j = 1$    | 76356  | 78552  | 28221 | 4473  | 324   | 6     |
| Outer  | (4, 2)     | 76880  | 83328  | 33268 | 6174  | 540   | 18    |
| Middle | $j = 2$    | 129156 | 123780 | 40644 | 5718  | 360   | 6     |
| Outer  | (4, 3)     | 107300 | 112980 | 44092 | 7938  | 648   | 18    |
| Outer  | (6, 3)     | 163620 | 158076 | 55080 | 8766  | 648   | 18    |

The generator `core_templates.py` independently enumerates all 29 labeled three-element preorders and all 82 compatible choices of the core vertices below the antichain, recovering these eleven polynomial forms. It symbolically counts roles with multiplicity  $\binom{m}{a}\binom{m-a}{b}$  for  $a$  antichain tails and  $b$  antichain heads. The layer checker processes antichain layers from bottom to top: if  $b$  unmatched earlier tails are available and  $k$  tails have been selected so far, choosing  $a$  new tails and  $c$  new heads is feasible exactly when  $c \leq b$ . It has multiplicity  $\binom{n_i}{a}\binom{n_i-a}{c}$  and updates  $(b, k)$  to  $(b + a - c, k + a)$ . Finishing with  $b = 0$  gives the support counts. This recurrence is compared with literal disjoint-support enumeration for all 96 compositions of six and seven.

### 3 Degree four (source 16, the base)

Source 16 is *Degree four preorder support polynomials* (17 pages as delivered), the four-part package `preorder-degree-four` (archives 16–19); it is the base of this report. Its files are shipped with the prefix `16-degree-four-`, except those that are byte copies of files of sources 20, 22 and 23, which are shipped once under those sources’ prefixes. Its abstract describes, among other things, the role-cover theorem of its Section 3, which is source 20’s article; see Section 3.3.

**Abstract of source 16, as delivered.** We study the polynomial counting disjoint directed matching supports of a finite preorder, with each feasible pair of endpoint sets counted once. We prove that the unweighted support polynomial of every finite preorder is rank-ultra-log-concave whenever its actual degree is at most four. The proof reduces arbitrary vertex counts to finitely many bounded cores with unbounded integer populations, and certifies the resulting inequalities by exact rational identities. Independently, we prove a structural theorem for every loopless directed relation whose arcs are covered by at most two selected tail vertices and two selected head vertices, with overlap allowed: its signed physical monomer polynomial is real stable, even with arbitrary nonnegative independent tail and head activities. Its support polynomial therefore has only negative real roots. The proof couples two rank-two transversal matroids through a stability-preserving operator and then merges the two copies of each physical vertex without allowing double use. For an unbalanced one-source, three-sink core, an ordinary proof of the last Newton inequality combines Lorentzian polarization with the Rayleigh property of rank-three matroids. A general exterior-only Lorentzian construction, an exact Gallai–Edmonds reduction, and independently checked rational certificates explain both the mathematical scope and the computational component. The final four-attachment ledger covers all 76 canonical templates; its last certificate consists of 2,273 rational square orbits and has been independently replayed. The result is computer-assisted, with ordinary structural proofs and finite algebraic verification distinguished throughout.

### 3.1 The support polynomial and the questions

A finite preorder is a reflexive transitive relation on a finite set  $V$ . Its off-diagonal relation is regarded as a directed graph  $D$ : an arc  $i \rightarrow j$  means  $i \neq j$  and  $i$  precedes  $j$ . Equivalent elements give opposite arcs. Several structural results below apply to directed relations without transitivity.

**Definition 3.1.** A feasible support of size  $k$  is an ordered pair  $(S, T)$  of disjoint subsets of  $V$ , each of cardinality  $k$ , for which there exists a bijection  $\pi : S \rightarrow T$  with  $s \rightarrow \pi(s)$  for every  $s \in S$ . Write

$$\gamma_k(D) = \#\{(S, T) : (S, T) \text{ is feasible of size } k\}, \quad \Gamma_D(t) = \sum_{k \geq 0} \gamma_k(D) t^k.$$

The empty support contributes  $\gamma_0 = 1$ . A support contributes once, regardless of the number of witnessing bijections.

The same-ground-set disjointness is essential. Passing to independent tail and head copies without subsequently excluding their simultaneous use changes this polynomial. Counting individual matchings also changes it.

For nonnegative tail activities  $u_i$  and head activities  $v_i$ , replace the contribution of  $(S, T)$  by

$$u^S v^T = \prod_{i \in S} u_i \prod_{j \in T} v_j.$$

We use the same notation when the weights are specified. A vertex may have different activities in its two roles. With zero activities, the polynomial degree can decrease.

Preorder polytopes and their lattice-point enumerators were introduced and studied by Chapoton and Athanasiadis [2]. The supplied ProveIt report identifies the unweighted support polynomial with its preorder gamma polynomial [1]. Our proofs concern the explicit support definition, so their validity does not require a particular formalization of that identification.

[Merge note: source 16's Lemma "degree" is Lemma 1.3, printed once in Section 1.3 with source 16's own proof.]

**Definition 3.2.** A polynomial  $g(t) = \sum_{k=0}^d g_k t^k$  is *rank-ultra-log-concave*, or rank-ULC, if its coefficients are nonnegative with no internal zeros and

$$\left( \frac{g_k}{\binom{d}{k}} \right)^2 \geq \frac{g_{k-1}}{\binom{d}{k-1}} \frac{g_{k+1}}{\binom{d}{k+1}} \quad (1 \leq k < d),$$

where  $d$  is the actual degree.

In degree four the inequalities are

$$3\gamma_1^2 \geq 8\gamma_2, \quad 4\gamma_2^2 \geq 9\gamma_1\gamma_3, \quad 3\gamma_3^2 \geq 8\gamma_2\gamma_4. \quad (3.1)$$

Normalization by the ground-set size, by a larger cover size, or by a presumed degree after some weights vanish is a different assertion.

[Merge note: source 16's Theorem "Universal first inequality" (if the underlying graph has matching number at most  $r \geq 2$ , then  $\gamma_1^2 \geq \frac{2r}{r-1} \gamma_2$ , including nonnegative role activities) is Theorem 1.5, printed once in Section 1.3 with source 16's Motzkin–Straus proof; it is *preorder-root-polytopes*, Theorem `mr:thm:first`.]

The preceding research established rank-ULC for every finite preorder of actual degree at most three by a separately auditable finite reduction (source 25, Theorem 2.3). It also exhibited the nonreal-rooted cubic  $1 + 24t + 162t^2 + 208t^3$ , whose discriminant is  $-2592$  (Section 2.9). Thus real-rootedness cannot be assumed in the general preorder problem.

### 3.2 An exterior-only Lorentzian construction

We use standard facts about Lorentzian polynomials: matroid basis polynomials are Lorentzian; products, nonnegative linear substitutions, coordinate upper truncations, and polarization preserve this class [6, 12]. For a homogeneous bivariate polynomial  $\sum a_k z^{r-k} t^k$ , the Lorentzian condition gives log-concavity of  $a_k / \binom{r}{k}$  with interval support.

**Theorem 3.3.** *Let  $P, Q, I$  be disjoint finite sets, and let the only arcs be  $P \rightarrow I$  and  $I \rightarrow Q$ . Put  $r = |P| + |Q|$ . With independent nonnegative exterior role activities, the polynomial*

$$H(z, t) = \sum_{k=0}^r F_k z^{r-k} t^k$$

*is Lorentzian, where  $F_k$  counts disjoint supports of size  $k$ . In particular  $F_k / \binom{r}{k}$  is log-concave. This is actual-degree ULC when  $\deg F = r$ .*

*Proof.* For the  $P$  side form a transversal matroid on the physical exterior vertices and private dummy elements  $d_p$ , one for each  $p \in P$ . Its presentation has right side  $P$ ; an exterior vertex has its given neighbors, and  $d_p$  is adjacent only to  $p$ . The dummies form a basis, so the rank is exactly  $|P|$ .

For  $P' \subseteq P$  and  $R \subseteq I$ , the set  $R \cup \{d_p : p \in P \setminus P'\}$  is a basis precisely when  $|R| = |P'|$  and  $R$  can be matched to  $P'$ . This encodes an endpoint support once, rather than counting its matchings. Construct the analogous  $Q$  matroid with distinct dummies and the same physical variables  $x_i$ .

Their squarefree basis polynomials  $B_P$  and  $B_Q$  are Lorentzian. Multiply them, then discard every monomial in which some  $x_i$  has exponent greater than one. Coordinate upper truncation preserves Lorentzianity [12, Proposition 3.3]. It removes exactly the choices using a physical exterior vertex in both roles. The all-dummy monomial survives.

Finally set every dummy variable equal to  $z$  and every physical variable equal to  $t$ . A surviving pair of bases specifies the support  $S = P' \cup L$ ,  $T = Q' \cup R$ , where  $L, R \subseteq I$  are disjoint. It contributes  $z^{r-|L|-|R|} t^{|L|+|R|}$ . This is exactly  $H$ . Exterior activities are introduced by scaling  $x_i$  separately in the two factors before multiplication. Every operation retains homogeneous degree  $r$ .  $\square$

**Remark 3.4.** If  $H$  has a common factor  $z$ , dividing that factor out is not a generally valid Lorentzian closure operation. The theorem does not silently strengthen its order- $r$  normalization to a smaller actual degree. It also does not assert real-rootedness.

For  $r = 4$ , differentiation of  $H$  twice gives the three constants in (3.1). For example  $\partial_t^2 H = 2F_2 z^2 + 6F_3 z t + 12F_4 t^2$ , whose discriminant yields  $3F_3^2 \geq 8F_2 F_4$ .

### 3.3 The two by two role cover

Source 16's Section 3, "Real-rootedness for a complete balanced core" (its Theorems "balanced", "twoedge", "missing" and "rolecover", its Lemmas "quadratic" and "asano", its Corollary "freeprod", the corollary on the gamma expansion, the interpolation remark, and its Appendix A of seven Rayleigh identities for two disjoint core edges), is source 20's article *Real stable monomers under a two by two role cover*: 176 of the 218 sentences of source 20 recur verbatim in source 16, with the same labels. It is printed once, from source 20, in Part II (Section on source 20; the main theorem is Theorem "Two by two role cover"). Source 16's text differs in four places, all printed there as marked second routes:

1. where source 20 invokes the stability-preserver theorem of Borcea and Brändén, source 16 proves each operator elementarily: an elementary Asano contraction lemma, a polarized auxiliary quadratic for the rank-two coupling, and the copy merge by diagonalization and differentiation;
2. the two-edge core covers canonical template 6, and the core with one missing edge covers canonical template 21, of the degree-four classification below;
3. the gamma-transfer corollary is stated with  $N = |V|$ ; source 20 allows any  $N \geq 2d$ ;
4. the appendix of source 20 has a typographical slip (“labels0,1”) that source 16 does not.

The degree-four ledger below discharges some templates by ordinary theorems, among them the two-edge core and the core with one missing edge (templates 6 and 21); the package records seven ordinary approvals, all for the last gap, of templates 1, 4, 6, 9, 10, 21 and 34.

### 3.4 The unbalanced one-source three-sink family

Let  $p$  be one source, let  $Q$  consist of three sinks, and let  $I$  be independent. Allow arbitrary arcs  $p \rightarrow I$  and  $I \rightarrow Q$ , and any subset  $p \rightarrow J$  of core arcs, where  $J \subseteq Q$ . No other arcs are present. All activities in this section are one.

**Theorem 3.5.** *Every relation in this family, and every reversal of such a relation, satisfies  $3\gamma_3^2 \geq 8\gamma_2\gamma_4$ .*

Let  $Q(t) = 1 + Lt + Bt^2 + Tt^3$  be the exterior-to-three-sink polynomial. Let  $H$  be the exterior neighbors of  $p$ ,  $m = |H|$ , and set

$$R(t) = \sum_{x \in H} Q_{-x}(t) = m + At + Ct^2 + Dt^3.$$

Let  $Q^+(t)$  be  $Q$  with one new exterior tail of neighborhood  $J$ . Then

$$\Gamma_D(t) = Q^+(t) + tR(t). \quad (3.2)$$

Indeed, supports with no exterior head belong to  $Q^+$ ; otherwise their unique exterior head  $x$  is matched from  $p$ , leaving exactly a  $Q_{-x}$  support. This excludes physical double use and introduces no matching multiplicity.

[Merge note: source 23 (Section 4) treats this family with the internal sink-side arc  $q_2 \rightarrow q_0$  added (canonical template 26), where Lemma 3.6 is not available, and replaces it by a deletion-mixture lemma.]

**Lemma 3.6.** *The homogeneous degree-three polynomial  $mz^3 + Az^2t + Czt^2 + Dt^3$  is Lorentzian. In particular  $C^2 \geq 3AD$ .*

*Proof.* Use the rank-three transversal matroid with private dummies. Identify dummy variables with  $z$ , the  $m$  variables indexed by  $H$  with  $s$ , and the other exterior variables with  $t$ . The resulting polynomial is Lorentzian. Polarize  $s$  into  $m$  slots using the degree bound  $m$  [6, Proposition 3.1]. Set one slot equal to zero and the other slots equal to  $t$ . A term using  $k$  elements of  $H$  is multiplied by  $\binom{m-1}{k} / \binom{m}{k} = (m-k)/m$ . Multiplication by  $m$  therefore gives the deletion sum. Vertices of  $H$  with no sink neighbor are matroid loops; they still count as unused polarization slots. If  $m = 0$  the assertion is immediate. No homogenizing monomial is divided out.  $\square$

**Lemma 3.7.** *For a three-center bipartite support polynomial, adjoining an exterior vertex cannot decrease the ratio of its cubic to quadratic coefficient, wherever that ratio is defined. Without division the assertion is  $T_{\text{new}}B - TB_{\text{new}} \geq 0$ .*

*Proof.* Every rank-three matroid is Rayleigh [14, Theorem 1.1]. Adjoin the new element  $e$  to the transversal matroid with three private dummy elements  $d_i$ . Let  $f$  be its basis polynomial. At existing exterior variables equal to one and all dummy variables and  $e$  equal to zero, we have

$$f = T, \quad \partial_e f = \Delta T, \quad \sum_i \partial_{d_i} f = B, \quad \sum_i \partial_e \partial_{d_i} f = \Delta B.$$

Summing the Rayleigh inequalities  $(\partial_e f)(\partial_{d_i} f) - f \partial_e \partial_{d_i} f \geq 0$  gives  $\Delta T B - T \Delta B \geq 0$ . Boundary evaluations follow by continuity. An added loop changes neither coefficient.  $\square$

*Proof of Theorem 3.5.* If  $D = 0$ , the desired inequality is immediate. Otherwise  $B, T, B^+, T^+, C$  are positive. For each deleted exterior vertex  $x$ , Lemma 3.7 applied when  $x$  is restored and when the new core tail is added gives  $T_x B \leq B_x T$  and  $T B^+ \leq B T^+$ . Thus  $T_x B^+ \leq B_x T^+$ , without dividing by a possibly zero  $B_x$ . Summing gives  $D B^+ \leq C T^+$ . Together with Lemma 3.6 and (3.2),

$$\begin{aligned} 3\gamma_3^2 - 8\gamma_2\gamma_4 &= 3(C + T^+)^2 - 8(A + B^+)D \\ &\geq \frac{C^2}{3} - 2CT^+ + 3(T^+)^2 = \frac{(C - 3T^+)^2}{3} \geq 0. \end{aligned}$$

$\square$

The real-rootedness conclusion of the balanced theorem does not extend to the full unbalanced family. For example the checked polynomial  $1 + 40t + 510t^2 + 2380t^3 + 2704t^4$  has discriminant  $-4112937216$  and two real roots. It is realized with four exterior vertices of each of the three two-sink neighborhoods, thirteen source-only exterior vertices, and all three core arcs. Thus there are twenty-five exterior vertices and twenty-nine vertices in total. The package contains the exact population specification and a direct support verifier.

### 3.5 A finite normal form for bounded matching rank

Let  $G$  be the comparability graph of a preorder of matching number  $r$ . In the canonical Gallai–Edmonds decomposition,  $D$  is the set of vertices missed by some maximum matching,  $A = N(D) \setminus D$ , and  $C = V \setminus (A \cup D)$ . The components  $D_i$  of  $G[D]$  are factor-critical,  $G[C]$  has a perfect matching, and

$$r = |A| + \frac{|C|}{2} + \sum_i \frac{|D_i| - 1}{2}. \quad (3.3)$$

These standard facts are proved, for example, by West [17].

**Proposition 3.8.** *Let  $I$  be the union of the singleton components of  $G[D]$ , and put  $K = V \setminus I$ ,  $a = |A|$ . Then  $I$  is independent, all its neighbors lie in  $A$ , and*

$$|K| \leq 3r - 2a.$$

*Moreover,  $\nu(G[K \setminus A]) = r - a$ . Every nonisolated exterior vertex is a singleton preorder equivalence class. Each attachment has one strict orientation toward all its exterior neighbors. Hence at most  $2^a - 1$  nonempty exterior types occur, with arbitrary integer populations.*

[Merge note: this proposition is source 25’s Theorem 2.6 (its first half), with the same proof: the matching budget (3.3) gives  $|K| \leq 3r - 2a$  and  $\nu(G[K \setminus A]) = r - a$ , and automorphism invariance of the canonical sets together with transitivity gives singleton exterior classes and one orientation per attachment (Lemma 2.4 there). Source 16’s four-paragraph proof is replaced by this pointer.]

For  $r = 4$  the core bounds are 10, 8, 6, 4 for  $a = 1, 2, 3, 4$ . When  $a = 0$ , each nontrivial component has at most nine vertices. Polynomials multiply over comparability components because every support splits uniquely. Rank-ULC is preserved by products: homogenize each factor at its actual degree, use the bivariate Lorentzian characterization and product closure, then specialize. This uses no change of normalization.

### 3.6 Exact support kernels and coverage

For a fixed core and a list of legal exterior types, let  $n_t$  be the population of type  $t$ . If  $q_t$  selected exterior labels are prescribed, let  $c_{k,q}$  count feasible supports in the induced relation on the core and precisely those prescribed labels, with every prescribed label required to occur. Then

$$\gamma_k(\mathbf{n}) = \sum_{|q| \leq \min(k,a)} c_{k,q} \prod_t \binom{n_t}{q_t}. \quad (3.4)$$

Every selected exterior vertex needs a different core partner, so at most  $a$  can occur. The binomial factors choose their physical labels. Feasibility depends only on the type quota and the selected core endpoint sets; unselected vertices cannot help match a fixed support. Thus (3.4) is an exact all-population formula, not interpolation from bounded numerical tests.

Independent kernel checks enumerate disjoint tail and head sets and apply all Hall inequalities to the induced directed bipartite relation. The producer uses a different Boolean matching routine. Both count one feasible support even when it has several witnesses.

Every preorder core is obtained from a quotient poset and positive sizes of its equivalence classes. Posets can be generated by adjoining a maximal element with every possible predecessor ideal. Canonicalization merges only complete adjacency matrices related by an explicitly tried permutation; global reversal exchanges tail and head supports and is also allowed. The four-core audit independently reconstructs the domain from upper-triangular quotient relations, block compositions, and all relabelings.

Exterior legality is checked with two identical copies of a type and with the union of all legal types. Any failed transitivity relation involves at most two exterior vertices, so these tests cover arbitrary multiplicities. For structural pruning, let  $H$  be a component of the graph formed with one clone of every legal type. Its largest matching rank over all its populations is

$$|A_H| + \nu((K \cap H) \setminus A_H),$$

where  $A_H$  is its set of active attachments. The upper bound follows by removing edges incident with  $A_H$ . Enough clones attain it by matching the attachments to distinct exterior vertices before matching the residual core. This is an all-population argument, not a finite sampling rule.

A connected bipartite preorder component with at least three vertices has no equivalence edge and no directed two-edge path on distinct vertices, since either would force a triangle. It therefore has genuine source and sink shores. Its support polynomial is the bipartite shore-support polynomial. The previously established weighted bipartite rank-at-most-four theorem applies [*Merge note*: source 16 cites here a weighted bipartite rank-four package that it embeds as a dependency; that package is batch-73 manuscript 26, which this intake prints in *matching-rank-normalization* as its source 77 (Theorem `mrn:v:rf4:thm:main`). The stronger *matching-rank-normalization*, Corollary `mrn:w:rf:cor:rankfive`, already covers every weighted bipartite graph of matching number at most five.]; an isolated two-element equivalence component contributes  $1 + 2t$ .

### 3.7 The bounded branches and the degree four theorem

All activities in the computational classification are one. The term *weighted quotient* below refers only to positive integer equivalence-class sizes, not to arbitrary role activities. The following

finite counts describe exact certificate domains, not random tests.

| Branch  | Core domain                    | Exact reduction                                       |
|---------|--------------------------------|-------------------------------------------------------|
| $a = 0$ | Components of order $\leq 9$   | Finite degree-four check and lower-degree theorem     |
| $a = 1$ | Marked core of order $\leq 10$ | 412622 coefficient pairs and 1237866 quadratic checks |
| $a = 2$ | Core of order $\leq 8$         | 89863 distinct gamma polynomials                      |
| $a = 3$ | Core of order $\leq 6$         | 6213 templates and 3437 distinct gamma polynomials    |
| $a = 4$ | Four attachment vertices       | 76 canonical nonbipartite templates                   |

The complete ten-element quotient scan checks 5049656 weighted quotient cases, 686481 of actual degree four. Smaller preorders are included by padding with isolated vertices. All actual-degree-four cases satisfy (3.1).

For  $a = 1$ , transitivity makes the attachment a singleton source or sink. Its  $m$  pendant clones give the exact identity

$$\Gamma_{K_m}(t) = \Gamma_K(t) + mt\Gamma_{K-v}(t).$$

Each gap is a quadratic  $Am^2 + Bm + C$ . The certificate verifies  $A, C \geq 0$  and either  $B \geq 0$  or  $B^2 \leq 4AC$ , proving nonnegativity for all real  $m \geq 0$ . Independent support enumeration reproduces all 412622 coefficient pairs and the complete quadratic ledger.

For  $a = 2$ , the core condition  $\nu(K \setminus A) \leq 2$  bounds the degree by four at every population. The three population variables have coefficient degree at most two. Exact enumeration produces 89863 distinct gamma polynomials and 179726 remaining-gap instances. Independent positivity replay covers 29215 normalized targets and all required face aliases.

For  $a = 3$ , 90042 legal marked specifications reduce structurally to 6213 templates and 3437 distinct gamma polynomials. Independent Hall recount verifies 188715 quota vectors and 943575 coefficients. All 204388 population faces are covered by exact rational identities, including 53722 normalized nontrivial targets.

For  $a = 4$ , independent enumeration gives 355 labeled four-element preorders, 4882 legal signed specifications, and 76 canonical retained templates. It checks 32516 quota vectors, 162580 coefficients, and the complete 258153-term gap data. These counts establish coverage and algebra before any positivity claim is made.

**Theorem 3.9** (Rank ultra log concavity through degree four). *For every finite preorder whose unweighted support polynomial  $\Gamma_D$  has actual degree  $d \leq 4$ , the sequence*

$$\left( \gamma_k / \binom{d}{k} \right)_{k=0}^d$$

*is log-concave. In particular, if  $d = 4$ , all three inequalities in (3.1) hold.*

*Proof.* Degrees at most three follow from the separately verified lower-degree theorem (source 25, Theorem 2.3). Suppose  $d = 4$ . The first inequality follows from Theorem 1.5. The normal form is exhaustive. The  $a \leq 3$  sectors satisfy all three gaps by the preceding exact certificates. For the  $a = 4$  sector, the complete independently replayed ledger covers both remaining gaps for all 76 templates and every nonnegative integer population, using the structural proofs above and the exact certificate schema below. Structural pruning and product closure handle discarded specifications and disconnected preorders. Every product is normalized by its own actual degree before homogeneous Lorentzian product closure is applied. Thus no vertex-count bound or unproved change of normalization enters the conclusion.  $\square$

Theorem 3.9 is computer-assisted. Its proof requires exact replay of a finite, explicitly supplied collection of identities; floating-point optimization is not a logical dependency. The final template, numbered 26, has core arcs  $2 \rightarrow 0$ ,  $3 \rightarrow 0$ ,  $3 \rightarrow 1$ ,  $3 \rightarrow 2$ , with vertex 3 the unique outward attachment. Its certificate contains 2273 nonnegative rational square orbits. Independent expansion verifies the full unscaled last-gap polynomial, thereby covering its 1785 remaining population faces. The complete ledger has no outstanding template or face obligations.

[Merge note: template 26 has a second, smaller route for its last gap: source 23, Section 4. The original route’s certificate (`global_integer_sextic_shifted_gmp_26.json`, 29.4 MB, the 2273 square orbits above) is not shipped with this report; its provenance record and replay log are (`data/16-degree-four-provenance-...`), and the README says how to retrieve the certificate from the arrival commit. It was replayed exactly at intake.]

### 3.7.1 A hybrid last-gap certificate

For a four-core template, delete all internal core arcs and let  $F_k$  be the exterior-only coefficients. Each size-four support uses all four core vertices on exterior edges, so  $\gamma_4 = F_4$ . Write  $B = \gamma_2 - F_2$  and  $C = \gamma_3 - F_3$ ; these count disjoint support classes using core edges and are nonnegative. Then

$$3\gamma_3^2 - 8\gamma_2\gamma_4 = (3F_3^2 - 8F_2F_4) + E, \quad E = 6F_3C + 3C^2 - 8BF_4. \quad (3.5)$$

The first summand is nonnegative by Theorem 3.3, at cover order four even if the exterior-only degree is smaller. For templates 9,36,37,50 the package proves  $E \geq 0$  by exact nonnegative binomial or binomial-square identities. Its degree is at most five, smaller than the full sextic gap. This is a case-specific reduction:  $E$  can be negative for other templates, and those cases use full-gap certificates instead.

## 3.8 Integer population certificates

Write  $B_\alpha(\mathbf{n}) = \prod_i \binom{n_i}{\alpha_i}$ . A global certificate has the form

$$P(\mathbf{n}) = \sum_j \frac{w_j}{|G|} \sum_{g \in G} B_{m_j}(g\mathbf{n}) \left( \sum_\alpha c_{j,\alpha} B_\alpha(g\mathbf{n}) \right)^2 + \sum_\alpha r_\alpha B_\alpha(\mathbf{n}), \quad (3.6)$$

where  $w_j, r_\alpha \geq 0$  are rational and  $G$  is a specified coordinate-permutation group. Each summand is nonnegative on  $\mathbb{Z}_{\geq 0}^d$ . These are identities of rational polynomials, but positivity is asserted on integer populations: for instance  $\binom{x}{2}$  is negative for  $0 < x < 1$ .

A standard-library checker expands every symmetry action and every product using

$$\binom{n}{a} \binom{n}{b} = \sum_{h=0}^{\min(a,b)} \frac{(a+b-h)!}{h!(a-h)!(b-h)!} \binom{n}{a+b-h}.$$

This counts pairs of subsets by their intersection size. Tensor products give the multivariate multiplication rule. Comparing every coefficient, including omitted zeros, proves the identity; checking all rational signs proves the inequality. Solver status and numerical tolerances are irrelevant to this final step.

For some smaller domains, split each coordinate into  $n_i = 0$  or  $n_i = 1 + y_i$ . These finitely many faces cover all integer populations. Each resulting polynomial is certified by nonnegative monomials and nonnegative monomial multiples of exact rational squares. Unused coordinates, permutations, and positive scaling factors are explicitly checked in the face ledger.

### 3.8.1 Shifted binomial discovery

The successful higher-dimensional search writes a square root accompanying  $B_m$  in the basis  $B_q(\mathbf{n} - m)$ . When an integer population is outside  $\mathbf{n} \geq m$ , the multiplier vanishes. Within that orthant, shifted coordinates reveal boundary zeros. The final output removes every shift using

$$\binom{n-m}{q} = \sum_{a=0}^q \binom{-m}{q-a} \binom{n}{a},$$

and is verified in the unshifted schema (3.6).

If a target vanishes on the entire lower integer box  $m + s$ ,  $0 \leq s \leq q$ , any square with multiplier  $B_m$  must vanish there. Triangular Newton interpolation then eliminates its coefficients indexed by  $a \leq q$ . An isolated zero imposes only its evaluation equation, not the deletion of a single basis coefficient. These restrictions aid discovery; only the final full identity is trusted.

### 3.8.2 Reproducing the finite verification

[Merge note: source 16’s package is not shipped in its delivered layout: its files are renamed into `code/` and `data/` with the prefix `16-degree-four-`, the files larger than 2 MB (the 93.7 MB three-attachment certificates among them) stay in the arrival commit, and the two embedded dependency archives are exact copies of source 25’s package and of batch-73 manuscript 26. The delivered `verify.py` first checks every one of its 260 manifest entries, so it runs only on the reconstructed package (README, “Reconstructing what is not shipped”). On Windows its path substitution writes backslashes into Python source; run it on Linux or WSL, or in a copy that uses `as_posix().`]

The accompanying directory `reproducibility/` contains the exact data, independent checkers, approved coverage inputs, and a portable driver. From that directory run

```
python verify.py --mode fast
```

This replays the fixed rational identities, gamma-to-gap reconstructions, face aliases, and complete template coverage. Its four-attachment manifest has 50 global identities: 24 middle-gap identities, 23 whole last-gap identities, and 3 identities for a core correction added to a Lorentzian baseline. They contain 15408 square orbits in total. Together with the ordinary proofs and coefficientwise-positive cases, the complete coverage check discharges all 39367 recorded outstanding face tasks and all 76 canonical templates.

For independent exhaustive kernel regeneration as well, run

```
python verify.py --mode full
```

The full mode adds the bounded-core support enumerations and Hall-based coefficient checks; its ten-element enumeration is the expensive step. Python 3.11 or later suffices for fixed certificate replay; full regeneration also requires GNU `g++` with C++17 support (the enumerators use GNU headers and 128-bit integers). Neither mode needs a numerical optimization package. The final package requires no partial-result override: a successful final coverage assertion demands all 76 templates and zero remaining tasks. A separate `--mode hashes` checks integrity only and does not establish the mathematics.

The previous degree-three and bipartite rank-four packages are included as explicit dependencies. The driver verifies their archive hashes; their own documented commands replay their earlier computational components. These are not silently re-enumerated by the new wrapper. Ordinary proof sources are supplied and pinned, but checking a source hash is likewise not a mathematical proof. The wrapper runs fresh copies of the independent checkers in a temporary

directory, removes historical positivity receipts, and derives its final coverage union from the newly replayed identities and the specified ordinary theorem approvals. The supplied README describes the small path-portability adaptations and where fresh logs and results are written.

### 3.9 Verification boundary and remaining research

The current article separates ordinary structural proofs from finite computer-assisted results. The role-cover theorem includes arbitrary independent role activities; the outward-star last-gap theorem is presently stated for unit activities. The exterior-only theorem has cover-order normalization, which is actual-degree normalization only at full cover rank. None is silently substituted for the general preorder theorem.

The completed degree-four ledger is included in the package. All 76 canonical four-attachment templates have both remaining gaps covered, and the lower-attachment branches and finite support kernels are independently checked. These finite identities, together with the ordinary reductions, establish Theorem 3.9 for arbitrary finite preorders of actual degree at most four. No Lean formalization or global priority claim is made here.

The most useful further questions are:

1. Replace the largest integer-population certificates, particularly the final one-source template with an internal sink-side edge, by ordinary structural proofs. Can its weighted deletion average be proved rank-ultra-log-concave directly?
2. Determine whether arbitrary independent role activities preserve rank-ULC for every degree-three preorder; a separate program proves the one-activity-per-vertex version and all unbounded branches for independent roles, but the remaining finite independent-role cases require their own coverage proof.
3. Determine whether actual-degree ULC holds for every two-by-three role cover. An unrestricted real-rootedness extension is impossible: the rank-three nonreal example, with two independent edges adjoined, already gives a nonreal degree-five product in that cover class.
4. Characterize when the exterior top-coefficient ratio remains monotone in ranks above three. General transversal matroids are not all Rayleigh, so the rank-three proof has a genuine boundary.
5. Classify equality in the degree-four inequalities and determine whether equality forces a factorization into lower-rank components.
6. Formalize the finite support kernel and the rational certificate checker in a proof assistant. This would strengthen the trust boundary without requiring formalization of the numerical discovery algorithm.
7. Find practical degree-five extensions of the bounded-core method, and identify the first true failures under actual-degree normalization rather than cover-size normalization.

[*Merge note:* status of these questions in this report. (1) Source 23 (Section 4) gives a smaller proof of the last gap of the final one-source template, by a deletion-mixture lemma, but it is still certificate-assisted (ten quartic identities) and does not prove the weighted deletion average rank-ULC directly. (2) Answered: source 22 (Part II) proves rank-ULC for every degree-three preorder with arbitrary independent role activities. (3) Open in general; settled for role covers with  $P \subseteq Q$  by source 11's corollary, for independent physical cores with saturation by source 15, and by real stability for two by two role covers (source 20); sources 11, 12 and 15 state the general two-tail/three-head case as open. (4)–(6) remain open. (7) Part III proves actual-degree five for five-vertex cores with a universal-sink cloud (source 34); degree five for preorders in general is open.]

## 4 A second route for the last gap of template 26 (source 23)

Source 23 is *A smaller proof of the last Newton gap for template 26. A certificate assisted supplement* (6 pages as delivered; archive `template26-structural-supplement`; no author line). Its files are shipped with the prefix `23-template26-`.

[*Merge note:* source 23 proves again, by a different and smaller route, the one entry of source 16's ledger with the largest certificate: canonical template 26, last gap (Section 3). Its family and the decomposition  $\Gamma = Q^+ + tR$  are those of source 16's Theorem 3.5 with the internal arc  $q_2 \rightarrow q_0$  added; its Rayleigh step and its final square are source 16's Lemma 3.7 and the last display of the proof of Theorem 3.5, with an internal-edge correction. It answers source 16's first research question in part. Its dependency "the independently approved vertex-weighted theorem at actual degree at most three" was delivered as archive 31, an earlier version of source 22; the statement is the case  $u_i = v_i$  of source 22's theorem, Corollary of source 22 in Part II. Both routes stay: source 16's certificate for template 26 is a direct binomial-square identity; source 23's route needs ten quartic identities, that theorem and Wagner's rank-three Rayleigh theorem.]

The text below is source 23's, from its opening paragraph.

This supplement proves the last order-four Newton inequality for the integer-population family called template 26. The argument uses a support decomposition, a deletion-mixture lemma, and monotonicity of a ratio of two top coefficients. Its finite algebraic input consists of ten exact quartic identities. The proof also depends on the independently approved vertex-weighted theorem at actual degree at most three [*Merge note:* that theorem is the case  $u_i = v_i$  of source 22's Theorem (Part II), recorded as Corollary of source 22.] and Wagner's rank-three Rayleigh theorem. Thus it is a smaller certificate-assisted proof, not a wholly ordinary proof.

### 4.1 Supports and the precise family

Let  $P$  be a finite preorder on a physical ground set  $V$ . A size- $k$  support is an *ordered* pair  $(U, W)$  of subsets of that same set such that

$$U \cap W = \emptyset, \quad |U| = |W| = k,$$

and some bijection  $U \rightarrow W$  uses off-diagonal arcs of  $P$ . Count each feasible ordered pair once, regardless of the number of matching witnesses. Distinct ordered pairs with the same union remain distinct. Set

$$\Gamma_P(t) = \sum_{k \geq 0} \gamma_k(P) t^k, \quad \gamma_0(P) = 1.$$

All activities in the template-26 conclusion are one. Diagonal preorder arcs do not contribute to supports because  $U$  and  $W$  are disjoint.

There are four core vertices  $q_0, q_1, q_2, p$ . Their off-diagonal arcs are

$$q_2 \longrightarrow q_0, \quad p \longrightarrow q_i \quad (0 \leq i \leq 2).$$

Exterior vertices are independent. An exterior vertex  $x$  may point to any subset  $N(x) \subseteq \{q_0, q_1, q_2\}$  satisfying

$$q_2 \in N(x) \implies q_0 \in N(x).$$

Independently, the arc  $p \rightarrow x$  may be present. In bit order  $q_0, q_1, q_2, p$ , the first three bits describe  $x \rightarrow q_i$ , while the fourth describes  $p \rightarrow x$ . The nonisolated types are exactly

$$1, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15. \tag{4.1}$$

Each type has an arbitrary nonnegative integer population. Isolates may be omitted. This is the canonical template with core rows  $[0, 0, 1, 7]$  and orientation mask 8.

**Theorem 4.1.** *For every such population, writing  $\Gamma(t) = \sum_k \gamma_k t^k$ , one has*

$$3\gamma_3^2 \geq 8\gamma_2\gamma_4. \quad (4.2)$$

*At actual degree four, this is the last Newton inequality for  $\gamma_k/\binom{4}{k}$ . If the degree drops below four, the displayed inequality is automatic.*

There can be at most three core heads and at most one exterior head, since only  $p$  points to exterior vertices. Thus the actual degree never exceeds four.

## 4.2 The support decomposition

Delete  $p$  and call the induced preorder  $Q$ . Its nonisolated exterior types are 1, 2, 3, 5, 7; let their populations be  $m = (a, b, c, d, e) \in \mathbb{Z}_{\geq 0}^5$ . A former type-8 vertex is now isolated. Write

$$Q(t) = 1 + At + Bt^2 + Tt^3.$$

Let  $H$  be the exterior vertices with  $p \rightarrow x$ , including the possible  $Q$ -isolates. For each  $x \in H$ , let  $Q_{-x}$  be the support polynomial after deleting that *physical vertex*, and put

$$R(t) = \sum_{x \in H} Q_{-x}(t) = \sum_{j=0}^3 R_j t^j.$$

Let  $Q^+$  be  $Q$  with one new exterior tail adjacent to all three core/head vertices. This is still a preorder, of support degree at most three. Write  $B^+ = [t^2]Q^+$  and  $T^+ = [t^3]Q^+$ .

**Proposition 4.2.** *The full support polynomial satisfies*

$$\Gamma(t) = Q^+(t) + tR(t). \quad (4.3)$$

*Consequently,*

$$\gamma_2 = B^+ + R_1, \quad \gamma_3 = T^+ + R_2, \quad \gamma_4 = R_3. \quad (4.4)$$

*Proof.* If the head set contains no exterior vertex, the support is exactly a support of  $Q^+$ , with its new universal tail identified with  $p$ . If it contains an exterior vertex  $x$ , then  $x$  is unique: only  $p$  can match to exterior heads. The pair  $p \rightarrow x$  is forced in every matching witness. Removing  $p$  and  $x$  gives precisely a support of  $Q_{-x}$ . Conversely, adjoining  $p \rightarrow x$  to such a support recovers a full support. Distinct  $x$  give distinct head sets. These are bijections of supports, so no matching multiplicity is introduced.  $\square$

For reference, the three coefficients of  $Q$  are the following polynomials:

$$A = 1 + a + b + 2c + 2d + 3e, \quad (4.5)$$

$$\begin{aligned} B = & ab + ac + ad + 2ae + bc + 2bd + 2be \\ & + 3cd + 3ce + 3de + \binom{c}{2} + \binom{d}{2} + 3\binom{e}{2} + b + c + e, \end{aligned} \quad (4.6)$$

$$\begin{aligned} T = & b \left( (d+e)(a+c) + \binom{d+e}{2} \right) \\ & + a \left( cd + ce + de + \binom{e}{2} \right) + \binom{c+d+e}{3} - \binom{c}{3} - \binom{d}{3}. \end{aligned} \quad (4.7)$$

Here binomial expressions are ordinary binomial polynomials, evaluated at integer populations. The accompanying independent Hall-support checker reconstructs these formulas directly: all 56 exterior quota vectors of total at most three are enumerated, and all 42 nonzero coefficient entries agree. The quota bound follows because every selected exterior vertex needs a distinct core head. The decomposition (4.3) is also independently replayed from direct ordered disjoint support counts, rather than assumed by the algebra checker.

### 4.3 Reducing deletion mixtures to pairs

For  $i \in \{a, b, c, d, e\}$  with  $m_i > 0$ , write  $Q_i = Q(m - e_i)$ , where  $e_i$  is the corresponding coordinate unit vector, and write  $Q_* = Q(m)$ . We require the following statement.

**Lemma 4.3** (Deletion mixtures). *Every nonnegative real linear combination  $S$  of  $Q_*$  and the available  $Q_i$  satisfies*

$$S_2^2 \geq 3S_1S_3. \quad (4.8)$$

*The populations  $m$  are nonnegative integers. In particular, (4.8) applies to  $R$  from Section 2: same-type deletions coincide, and deletion of a  $Q$ -isolate contributes  $Q_*$ .*

**The two-point principle.** Consider any finite list of triples  $(A_i, B_i, T_i)$  with  $A_i > 0$ . For a nonzero nonnegative mixture, normalize by  $A = \sum_i w_i A_i$  and let  $\lambda_i = w_i A_i / A$ . Then

$$\left(\frac{B}{A}, \frac{T}{A}\right) = \sum_i \lambda_i \left(\frac{B_i}{A_i}, \frac{T_i}{A_i}\right), \quad \lambda_i \geq 0, \quad \sum_i \lambda_i = 1.$$

At each fixed horizontal coordinate, a highest point of this finite convex hull is a combination of at most two listed points. To see this, take a maximizing combination with smallest positive support. If it has at least three positive coefficients, a nonzero perturbation  $\delta$  satisfies  $\sum \delta_i = \sum \delta_i (B_i/A_i) = 0$ . Both sufficiently small signs are feasible. Maximality forces  $\sum \delta_i (T_i/A_i) = 0$ ; extend one sign until a coefficient vanishes, contradicting minimality. This also covers repeated horizontal coordinates.

It follows that every mixture satisfies  $B^2 \geq 3AT$  if every two-point mixture does: all upper boundary points then lie below  $y = x^2/3$ , as does the entire hull. This does *not* assume that the subgraph of this convex parabola is convex. The zero mixture is immediate. In our family every  $A_i \geq 1$ , because the internal arc remains.

**Undeleted and deleted pairs.** Give one marked physical exterior vertex  $x$  of type  $i$  activity  $s \in [0, 1]$ , keeping every other vertex activity equal to one. A support has weight equal to the product of the activities of its physical vertices. Therefore its polynomial is exactly

$$Q^{(s)}(t) = sQ_*(t) + (1-s)Q_i(t). \quad (4.9)$$

Indeed, a support either uses  $x$  once or avoids it. The independently approved vertex-weighted actual-degree-at-most-three theorem [*Merge note*: that theorem is the case  $u_i = v_i$  of source 22's Theorem (Part II), recorded as Corollary of source 22.] gives  $B(s)^2 \geq 3A(s)T(s)$  when the surviving degree is three. At lower degree  $T(s) = 0$ , so the same displayed inequality is automatic, including  $s = 0$ . This is a change in one vertex's activity, not a substitution of a fractional population into (4.5)–(4.7).

For  $u, v \geq 0$  with  $u + v > 0$ , choose  $s = u/(u + v)$  and multiply (4.9) by  $u + v$ . Homogeneity gives the inequality for  $uQ_* + vQ_i$ . The same theorem supplies every endpoint gap  $B_i^2 - 3A_iT_i \geq 0$ . It remains to check the ten distinct-deletion pairs.

### 4.4 The ten exact quartic identities

Write  $(A_i, B_i, T_i)$  for the coefficients of  $Q_i$  and define

$$C_{ij}(m) = 2B_iB_j - 3(A_iT_j + A_jT_i) \quad (i < j).$$

Since both deletion types are available, write  $m = x + e_i + e_j$  with  $x \in \mathbb{Z}_{\geq 0}^5$ . Thus the two coefficient triples come from  $Q(x + e_j)$  and  $Q(x + e_i)$ .

**Lemma 4.4** (Computer-checked quartic identities). *For each of the ten unordered pairs below, the supplied exact data give an identity*

$$C_{ij}(x + e_i + e_j) = \sum_{\ell} q_{\ell} \binom{x}{\alpha_{\ell}} f_{\ell}(x)^2 + \sum_{\beta} r_{\beta} \binom{x}{\beta}, \quad (4.10)$$

where  $q_{\ell}, r_{\beta} > 0$  are rational,  $\binom{x}{\alpha} = \prod_{k=1}^5 \binom{x_k}{\alpha_k}$ , all exponent vectors belong to  $\mathbb{Z}_{\geq 0}^5$ , and each  $f_{\ell}$  is an explicitly listed rational polynomial in the same binomial basis. Consequently  $C_{ij}(m) \geq 0$  on its integer population domain.

| Deleted population coordinates | Square terms | Remainder terms |
|--------------------------------|--------------|-----------------|
| $a, b$                         | 15           | 84              |
| $a, c$                         | 21           | 81              |
| $a, d$                         | 18           | 83              |
| $a, e$                         | 22           | 83              |
| $b, c$                         | 24           | 81              |
| $b, d$                         | 19           | 88              |
| $b, e$                         | 21           | 85              |
| $c, d$                         | 18           | 89              |
| $c, e$                         | 19           | 86              |
| $d, e$                         | 19           | 86              |
| Total                          | 196          | 846             |

*Certificate format and verification.* The data file `distinct_deletion_cross_certificates.json` records, for each pair, its deletion indices ( $0, \dots, 4$  correspond to  $a, \dots, e$ ), the population shift, and the two sums in (4.10). A square record contains a rational weight, a multiplier exponent vector, and a list of exponent-coefficient pairs specifying  $f_{\ell}$ . A remainder record is an exponent-coefficient pair. All ten identities are checked coefficientwise with exact rational arithmetic, together with every positivity and integer-exponent condition. Each summand is nonnegative for  $x \in \mathbb{Z}_{\geq 0}^5$ . Integer scope matters: ordinary binomial polynomials need not be nonnegative at arbitrary nonnegative real arguments.

The independent checker reconstructs the targets from Hall support feasibility and imports no producer code. It also derives binomial multiplication constants by counting pairs of subsets with prescribed union. Thus the finite input is an exact identity certificate, rather than a sample of population values or a floating-point optimization result.

*Completion of Lemma 4.3.* Write  $G(S) = S_2^2 - 3S_1S_3$ , so  $G_i = G(Q_i) \geq 0$ . For  $u, v \geq 0$ ,

$$G(uQ_i + vQ_j) = u^2G_i + uvC_{ij} + v^2G_j \geq 0.$$

Lemma 4.4 settles all distinct-deletion pairs. Section 3 settles the undeleted/deleted pairs; identical deletion types need no extra check. The two-point principle proves (4.8), hence  $R_2^2 \geq 3R_1R_3$ .  $\square$

## 4.5 The top ratio with the internal edge present

Remove  $q_2 \rightarrow q_0$  from  $Q$  and call the exterior-only support polynomial  $F$ . Set  $B_0 = F_2$ ,  $T_0 = F_3$ , and let  $L$  be the number of exterior vertices adjacent to  $q_1$ . At support level,

$$B = B_0 + L, \quad T = T_0. \quad (4.11)$$

A size-two support using the internal edge must match its sole exterior tail to  $q_1$ , because  $q_2$  is a tail and  $q_0$  a head. A size-three support cannot use the internal edge: it would require three core heads while  $q_2$  is already a tail.

Adjoin one exterior vertex  $x$  of any allowed outgoing neighborhood, possibly empty. Set

$$\beta = \Delta B_0, \quad \tau = \Delta T_0, \quad \ell = \Delta L \in \{0, 1\}.$$

We prove the division-free top-ratio inequality

$$T_{\text{new}}B - TB_{\text{new}} = \tau(B_0 + L) - T_0(\beta + \ell) \geq 0. \quad (4.12)$$

**Exterior-only contribution.** Form the transversal matroid whose elements are the old exterior vertices,  $x$ , and three private dummies  $d_0, d_1, d_2$ , each  $d_i$  adjacent only to  $q_i$ . Its independent sets are those matchable injectively into the three heads; the dummies ensure rank exactly three. Let  $f$  be its multiaffine basis polynomial, counting each feasible basis once. Wagner's theorem [14] gives, for distinct elements  $e, h$ ,

$$f_e f_h - f_{eh} \geq 0$$

for positive weights and hence, by continuity, for nonnegative weights. Subscripts here mean partial derivatives.

Evaluate with all old exterior weights one and the weights of  $x, d_0, d_1, d_2$  zero. Then

$$f = T_0, \quad f_x = \tau, \quad \sum_{i=0}^2 f_{d_i} = B_0, \quad \sum_{i=0}^2 f_{x d_i} = \beta.$$

For example, a basis using  $d_i$  corresponds to an exterior two-tail support on the complementary pair of heads; that omitted head is unique. This preserves support counting even when a tail pair admits several choices of head pair. Summing the three Rayleigh inequalities for  $x, d_i$  gives

$$\tau B_0 - T_0 \beta \geq 0. \quad (4.13)$$

**Contribution of the internal edge.** Let  $K$  count exterior two-element supports on the fixed head pair  $\{q_0, q_2\}$ . Then

$$T_0 \leq LK. \quad (4.14)$$

For every feasible exterior triple choose one matching witness, and record the vertex assigned to  $q_1$  and the remaining pair. These data determine the triple, giving an injection into the pairs counted by  $LK$ . If  $\ell = 1$ , adjoining  $x$  to each pair counted by  $K$  creates a distinct feasible triple, so  $\tau \geq K$  and  $\tau L - T_0 \ell \geq LK - T_0 \geq 0$ . If  $\ell = 0$ , this expression is simply  $\tau L \geq 0$ . Adding this to (4.13) proves (4.12). An isolated new vertex gives equality. No positivity or division is needed here.

## 4.6 Compatibility and the final square

If  $R_3 = 0$ , then (4.4) gives  $\gamma_4 = 0$ , so (4.2) is immediate. Assume  $R_3 > 0$ . Some  $Q_{-x}$  has a positive cubic coefficient. Restoring a vertex preserves its supports; restricting a matching witness gives a quadratic support. Hence  $T > 0$  and  $B > 0$ .

For  $x \in H$ , write  $B_x = [t^2]Q_{-x}$  and  $T_x = [t^3]Q_{-x}$ . Apply (4.12) first when restoring  $x$ , then when adjoining the universal exterior vertex that defines  $Q^+$ :

$$T_x B \leq B_x T, \quad TB^+ \leq BT^+.$$

Multiply by nonnegative factors to obtain

$$T_x B B^+ \leq B_x T B^+ \leq B_x B T^+.$$

Cancel only the strictly positive factor  $B$ . Summing over  $x \in H$  yields

$$R_3 B^+ \leq R_2 T^+. \quad (4.15)$$

The argument covers  $B_x = 0$  or  $T_x = 0$  without division by either. An isolate deletion gives equality in the first comparison.

By Lemma 4.3,  $R_1 R_3 \leq R_2^2/3$ . Combining this with (4.4) and (4.15) proves

$$\begin{aligned} 3\gamma_3^2 - 8\gamma_2\gamma_4 &= 3(R_2 + T^+)^2 - 8(R_1 + B^+)R_3 \\ &\geq 3(R_2 + T^+)^2 - \frac{8}{3}R_2^2 - 8R_2T^+ \\ &= \frac{1}{3}(R_2 - 3T^+)^2 \geq 0. \end{aligned}$$

This completes the proof of Theorem 4.1.

## Scope and reproducibility

This supplement proves the last gap. Together with the separately approved middle-gap proof and general first-gap bound, it gives the smaller proof for this family. The original degree-four theorem and frozen certificate archive are unchanged. There is no claim of a general weighted degree-four theorem, fractional-population theorem, real-rootedness, or novelty.

[*Merge note:* the middle-gap proof meant here is source 16’s certificate for template 26, gap two (shipped as `data/16-degree-four-a4-gram_z26.json`), and the first-gap bound is Theorem 1.5. The embedded `weighted-degree-three-package.zip` is an exact copy of archive 31, superseded by source 22, and is not shipped; because source 23’s `verify.py` checks it against its checksum list, the delivered wrapper runs only on the reconstructed package (README). The same Windows path problem as in source 16’s wrapper applies.]

The tradeoff is explicit: ten quartic certificates replace a much larger direct last-gap certificate, using the already approved vertex-weighted degree-three theorem. The supplement includes the certificates, independent exact checkers, approval receipts, and `weighted-degree-three-package.zip`. From the extracted supplement root, run `python verify.py` for the two local independent replays and the pinned dependency archive. Run `python verify.py --verify-dependency` to extract and replay the full portable verifier of that dependency as well. The local replays check the exact quartic certificates and the support decomposition coefficientwise. The full mathematical assembly has independent approval, including the convex-hull argument, degree-drop endpoints, internal-edge correction, zero cases, and final square.

# Appendix

## A Provenance and merge record

### A.1 The archives

All eighteen archives arrived in ProveIt commit `f8c3a392a` on 1 October 2026 and were placed in commit `8f5c53106` (batch 73, cluster C2); archives 11, 12 and 15 were handed over to this

cluster by the cluster that placed batch-73 manuscript 26 in *matching-rank-normalization*. Only sources 16 and 25 cite a ProveIt commit: `d5e863bba` (*preorder-root-polytopes* at “Write batch 70 (4/5)”); the labels they cite are unchanged since. All other 40-digit hexadecimal strings in the packages are SHA-256 digests of package files. Every package’s checksum ledger verified at intake.

| Src. | Archive(s)                                                                                               | Manuscript (pages as delivered)                                                         | Pin                    | Printed             |
|------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------|---------------------|
| 25   | <code>preorder-gamma-degree-three-package</code>                                                         | Rank ultra log concavity for preorder gamma polynomials of degree at most three (14)    | <code>d5e863bba</code> | Section 2           |
| 16   | archives 16–19, <code>preorder-degree-four-part-k-of-4</code> , $k = 4, 3, 2, 1$                         | Degree four preorder support polynomials (17); the base                                 | <code>d5e863bba</code> | Section 3           |
| 23   | <code>template26-structural-supplement</code>                                                            | A smaller proof of the last Newton gap for template 26 (6)                              | —                      | Section 4           |
| 20   | <code>role-cover-package</code>                                                                          | Real stable monomers under a two by two role cover (9)                                  | —                      | Section in Part II  |
| 12   | <code>three-core-rayleigh-result</code>                                                                  | Coefficientwise Rayleigh inequalities for a three vertex shore (6)                      | —                      | Section in Part II  |
| 11   | <code>physical-cover-three-result</code>                                                                 | Rank ultra log concavity under a three vertex physical cover (5)                        | —                      | Section in Part II  |
| 15   | <code>bipartite-and-pendant-support-result</code>                                                        | Lorentzian and real stable directed support polynomials (6)                             | —                      | Section in Part II  |
| 22   | <code>independent-role-degree-three-package</code>                                                       | Independently weighted preorder support polynomials of actual degree at most three (21) | —                      | Section in Part II  |
| 14   | <code>universal-sink-last-gap-package</code>                                                             | Sharp final coefficient bounds for directed cores with universal sinks (4)              | —                      | Section in Part III |
| 10   | <code>universal-sink-all-degrees-package</code>                                                          | Rank ultra log concavity for four vertex cores with universal sinks (5)                 | —                      | Section in Part III |
| 34   | <code>universal-sink-five-core-package</code>                                                            | Rank five ultra log concavity for five vertex cores with universal sinks (6)            | —                      | Section in Part III |
| 06   | archives 07 and 06, <code>universal-sink-five-core-boundary-package-part-1</code> , <code>-part-2</code> | The rank four boundary for five vertex cores with universal sinks (7)                   | —                      | Section in Part III |
| 31   | <code>weighted-degree-three-package</code>                                                               | Vertex-weighted preorder support polynomials of actual degree at most three (17)        | —                      | superseded by 22    |
| 13   | <code>universal-sink-degree-four-package</code>                                                          | Weighted degree four rank ultra log concavity for universal sink extensions (5)         | —                      | superseded by 10    |

## A.2 What each source contributed

- 25: unit-activity rank-ULC through actual degree three; the bounded-core normal form at every degree; the minimum order fifteen of a nonreal-rooted degree-three gamma polynomial and its uniqueness; the ordinal-sum family. Its first inequality is *preorder-root-polytopes*’s.
- 16 (base): unit-activity rank-ULC at actual degree four; the exterior-only Lorentzian construction; the outward-star last gap; the kernel formula and coverage, the bounded branches

and the integer-population certificate schema. Its Section 3 and Appendix A are source 20's article (printed from source 20); its first inequality is *preorder-root-polytopes*'s.

- 23: a second, smaller route for the last gap of template 26.
- 20: real stability of the signed monomer polynomial under every two by two role cover, with independent activities; the free-product corollary; the gamma transfer. It generalizes *matching-rank-normalization*, Theorem `mrn:w:sb:thm:main` (the bipartite case with disjoint cover sides) by a different proof.
- 12: the coefficientwise three-core boundary inequality (computer-assisted), the full coefficientwise Rayleigh property, negative correlation, and a four-core obstruction.
- 11: rank-ULC under every physical vertex cover of size at most three, with arbitrary internal core arcs; bipartite strictness and equality.
- 15: the bipartite physical Lorentzian theorem, overlapping role covers with an independent core, the pendant-head stability criterion and its elementary seed.
- 22: rank-ULC of every finite preorder of actual degree at most three with independent role activities; the articulation theorem; the zero-activity role filter; the half-plane-property side-matroid criterion and the final rank-32 Gram certificate. Its three-core and one-tail theorems are special cases of sources 11 and 15. It supersedes archive 31.
- 14: the sharp last-coefficient bound  $K(r, m)$  for every core size with universal sinks.
- 10: rank-ULC at every actual degree for four-vertex cores with universal sinks, through the certified inequality  $\gamma_2^2 \geq 3\gamma_1\gamma_3$ . It supersedes archive 13.
- 34: rank-ULC at actual degree five for five-vertex cores with universal sinks, and a counterexample to the constant three there.
- 06: the actual-degree-four boundary of the five-vertex family, and the strengthened ordinary core comparisons.

### A.3 Where the merge had to choose

1. *One report, three Parts.* All sources answer *preorder-root-polytopes*'s Research question `mr:q:gamma`; they would be additions to *preorder-root-polytopes*, which already has thirteen Parts and whose batch-72 continuation was already split off into *matching-rank-normalization*. They form one new report beside *matching-rank-normalization*.
2. *Base.* Source 16, the farthest-reaching unit-activity result. Its Section 3 and Appendix A are printed once, from source 20, in Part II; source 16's different operator proofs are kept there as second routes (Section 3.3).
3. *Common definitions and the first inequality* are printed once (Section 1.3); the sources' copies are pointers.
4. *Same theorem, two sources.* Source 25's theorem is the unit case of source 22's: both proofs are printed, in Parts I and II. Source 23 re-proves one entry of source 16's ledger: both routes are described, source 23's printed. Source 14's sharp bound contains the last inequalities of sources 10 and 34 (and of archive 13): it is printed once, and those proofs, which are the same route at  $r = 4, 5$ , are pointers. Sources 11 and 15 contain source 22's three-core and one-tail theorems: source 22's three-core proof, an ordinary argument through Wagner's rank-three theorem, is kept as a second route; its one-tail proof is the pendant proof of source 15 and is a pointer.

The three-squares identity of sources 11, 12 and 22 is printed once (in source 12's section); the moment envelope of sources 10, 34 and 06 once (in source 10's); the core comparisons of sources 34 and 06 once, in source 06's stronger form; the nonreal cubic of sources 16, 22, 25 and archive 31 once (in source 25's).

5. *Superseded versions.* Archive 31 (vertex activities) is an earlier version of source 22: same skeleton, eleven byte-identical proof notes; its theorem is printed as the case  $u_i = v_i$  of source 22's. Its only own material, a ledger of vertex-only certificates for a finite special case, survives in the arrival commit. Archive 13 is an earlier version of source 10 with a weaker middle inequality,  $4\gamma_2^2 \geq 9\gamma_1\gamma_3$ , certified by binomial squares only, and an alternative first-gap argument valid in degree four; it survives in the arrival commit.
6. *Source numbers* of multi-archive packages are their lowest archive numbers (16 and 06), as are their file prefixes.
7. *Not shipped* (README): embedded exact copies of other archives, PDFs, delivery READMEs, checksum ledgers, rendered page images, build logs and scripts of the delivered PDFs, the files of source 16 larger than 2 MB, source 06's two bounded-sink certificate archives and their manifest, and the regenerable outputs listed in the README. Source 34's 9,610 certificate files are shipped as one derived container whose members are byte-identical to the delivered files.

## A.4 Shipped files

Each source's code, data, audits and proof notes are shipped as delivered, renamed with its prefix (25-degree-three-, 16-degree-four-, 23-template26-, 20-role-cover-, 12-three-core-rayleigh-, 11-physical-cover-, 15-pendant-support-, 22-independent-role-, 14-sink-last-gap-, 10-sink-all-degrees-, 34-five-core-, 06-five-core-boundary-): code in `code/`, data and records in `data/`, audits and proof notes at the top of the report directory. The report README lists them, maps the delivered names, lists what is not shipped and how to retrieve it from the arrival commit, and gives rerun instructions that work on copies. The text of delivered files still uses delivered paths.

## References

- [1] ProveIt project report, *A Reflexive Root-Polytope Model for Preorder  $h$ -Polynomials* (*preorder-root-polytopes*), cited by sources 16 and 25 at the pinned source revision d5e863bba6b40738af292b45005a0cf0495903e4, Parts IV and XII. <https://github.com/VladimirReshetnikov/ProveIt/blob/d5e863bba6b40738af292b45005a0cf0495903e4/SetTheory/Cardinals/docs/reports/enumerative-combinatorics/preorder-root-polytopes/article.tex>. Current version: SetTheory/Cardinals/docs/reports/enumerative-combinatorics/preorder-root-polytopes/.
- [2] C. A. Athanasiadis and F. Chapoton, *Polytopes and posets associated to preorders*, arXiv:2605.26916v1 (2026), Section 5 and Proposition 9.2. <https://arxiv.org/abs/2605.26916>. (Source 16 lists the authors as F. Chapoton and C. A. Athanasiadis.)
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- [9] M. Kummer and B. Sert, *Matroids on Eight Elements with the Half-plane Property and Related Concepts*, arXiv:2111.09610v4, Sections 5 and 6, especially Theorem 5.2. <https://arxiv.org/abs/2111.09610>.
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